Greek Corpus Pipeline: diagnosing annotation variation in Ancient Greek dependency treebanks

Supplementary materials

These materials document the analyses reported in the article. They consist of eleven sections, S1–S11. The article cites sections and subsections as S8.1, S10.6 and so on, and tables as Table S2.1, Table S3.2 and so on; sections and tables are numbered in two separate series, so that, for example, section S10.6 contains Table S10.5. References without the prefix S are to the article. All the results come from six runs of Greek Corpus Pipeline 3.1.0, one for each of the six strata examined — Gorman, Pedalion and the automatic and manual parts of the two GLAUx samples — identified with their inputs in S11. The comparison with Stanza and UDPipe (§ 7 of the article) was carried out within the two runs on Gorman and on Pedalion.

The sections are the following. S1 gives the catalogue of the 77 checks with their counts; S2 the specification of each check; S3 the authority on which each criterion rests; S4 the nine numerical parameters; S5 the sensitivity analysis of the thresholds; S6 the ten closed lists; S7 ten case studies; S8 the comparison with the variation n-gram; S9 the two GLAUx samples; S10 the comparison with the two automatic analysers; S11 the counting units, the frozen results, the provenance of the data and what is needed to repeat the measurements.

S1. The complete catalogue of the 77 checks

Table S1.1 lists the 77 checks with their family, diagnostic class and treatment (§ 4.1 of the article) and the number of findings in each of the six strata, 160,577 in all. The columns of the strata are, in order: Gorman, Pedalion, the automatic and the manual stratum of the first GLAUx sample (called the development sample in the abstract of the article), and the automatic and the manual stratum of the confirmation sample; hereafter the two samples are called “first sample” and “confirmation sample”. A dash means that the check found no occurrence of its phenomenon in that stratum. The only check that cannot fire at all with the filter adopted in the study is nonlexical-z-pos (see below). Table S1.2 sums the findings by class and by stratum, and Table S1.3 shows in which treebanks each check fires.

The same 77 checks are listed in the file data/catalogo_controlli.csv distributed with the program, with the same identifiers, names, families, classes and treatments. The file also contains what the program needs in order to run them, such as the severity shown in the interface and the texts of the findings in Italian and English; it does not contain the authority of the criterion, which is given in S3. The specifications of S2 use the terminology of the article.

Table S1.1 The 77 checks, ordered by diagnostic class and, within each class, by family and by identifier. Class: V = verifiable violation, S = systematic variant, C = case to check. Treatment: A = automatic correction, M = manual review.

Identifier	Name	Family	Cl	Tr	Gorman	Pedal.	GL auto.	GL man.	Conf. auto.	Conf. man.
creon-anomaly	Nominal χρεών with an incomplete PoS	morphology	V	M	1	—	—	—	—	—
dein-participle	δεῖν with participial mood	morphology	V	A	36	—	—	—	—	—
deina-anomaly	Pronominal δεῖν’ with an incomplete PoS	morphology	V	M	2	—	—	—	—	—
invalid-character	Character not allowed in the morphological slot	morphology	V	M	36	62	—	—	—	—
invalid-length-broken	Label of incorrect length to be reconstructed in context	morphology	V	M	—	16	—	—	—	—

Identifier	Name	Family	Cl	Tr	Gorman	Pedal.	GL auto.	GL man.	Conf. auto.	Conf. man.
invalid-length-padding	Label consisting only of hyphens with an incorrect length	morphology	V	A	320	79	—	—	—	—
leading-t	PoS not allowed in slot one (t)	morphology	V	M	6	—	—	—	—	—
leading-x-lexical	PoS x on a lexical record	morphology	V	M	18	—	—	—	—	—
missing-postag	PoS tag missing	morphology	V	M	26	33	—	—	—	—
numeral-generic-m	Numeral with underscores in the PoS tag	morphology	V	A	24	4	—	—	—	—
osfranthiti-anomaly	ὀσφράνθητι with an incompatible morphological analysis	morphology	V	A	2	—	—	—	—	—
participle-mood-g	Value not allowed in the mood slot (g)	morphology	V	M	8	—	—	—	—	—
pot-anomaly	Elided forms πότ’/πόθ’ with an adjectival PoS	morphology	V	M	2	—	—	—	—	—
punctuation-leading-e	PoS e on a punctuation mark	morphology	V	A	17	—	—	—	—	—
san-anomaly	σὰν as a letter name with an incomplete PoS	morphology	V	M	1	—	—	—	—	—
slot-forbidden	Feature not allowed for the part of speech	morphology	V	M	34	19	—	—	—	—
slot-missing	Required morphological feature left unfilled	morphology	V	M	453	353	356	9	128	—
tessereskaideka-anomaly	τεσσερεσκαίδεκα with a non-numeral PoS	morphology	V	A	4	—	—	—	—	—
missing-lemma	Lemma missing	lemmatisation	V	M	150	42	—	—	—	—
apos-suffix-orphan	The suffix _AP without an apposition	syntactic structure	V	M	56	—	—	—	—	—
coord-suffix-orphan	The suffix _CO without a coordinating node	syntactic structure	V	M	442	440	—	8	—	9
head-cycle	Cycle in the heads	syntactic structure	V	M	—	23	—	2	—	2
head-dangling	Non-existent head	syntactic structure	V	M	—	21	9	—	21	—
head-missing	Syntactic head not declared	syntactic structure	V	M	22	15	—	—	—	—
relation-greek-lookalike	Look-alike Greek character in the syntactic label	syntactic structure	V	A	10	—	—	—	—	—
relation-missing	Relation missing	syntactic structure	V	M	190	46	—	—	—	—
relation-nil-placeholder	The placeholder nil in place of the relation	syntactic structure	V	M	3	—	—	—	—	—
relation-truncated-or-foreign	Non-AGDT relation traceable to a known convention	syntactic structure	V	M	20	11	—	—	—	1
relation-typo	Unambiguous typo in the syntactic label	syntactic structure	V	A	1	—	—	—	—	—
relation-unknown	Relation outside the scheme	syntactic structure	V	M	2	57	9	22	23	—
sentence-no-root	Sentence without a root	syntactic structure	V	M	6	32	—	—	—	3
sentence-not-annotated	No head declared in the sentence	syntactic structure	V	M	—	3	—	—	—	—
token-id-duplicate	Duplicate identifier	syntactic structure	V	M	—	12	—	—	—	—
annotator-note-in-relation	Annotator’s note in the relation field	text and encoding	V	M	—	43	—	—	—	—
beta-code-form	Beta code residues in the form	text and encoding	V	M	—	—	—	—	—	—
beta-code-lemma	Beta code residues in the lemma	text and encoding	V	M	—	1	—	—	—	—
breathing-impossible	Elision apostrophe encoded as a breathing	text and encoding	V	A	19	—	—	—	—	—
final-sigma-inside	Final sigma inside the word	text and encoding	V	A	1	2	—	—	—	—
invisible-char-form	Format characters in the form	text and encoding	V	A	—	—	—	—	—	—

Identifier	Name	Family	Cl	Tr	Gorman	Pedal.	GL auto.	GL man.	Conf. auto.	Conf. man.
invisible-char-lemma	Format characters in the lemma	text and encoding	V	A	—	—	—	—	—	—
isolated-breathing	Isolated breathing in the form	text and encoding	V	M	—	31	—	—	—	—
mixed-script	Look-alike Latin letters in a Greek word	text and encoding	V	A	—	2	—	—	—	—
mixed-script-unrepairable	Latin letters that are not look-alikes in a Greek word	text and encoding	V	M	—	11	3	—	—	—
whitespace-in-form	Internal separator in the form	text and encoding	V	M	1	1	—	—	—	—
whitespace-in-lemma	Internal separator in the lemma	text and encoding	V	M	1	2	1	—	—	—
degree-underscore	Underscore in the degree slot	morphology	S	A	11,187	270	—	—	—	—
grade-p	Positive degree marked with p	morphology	S	A	1,218	454	—	—	—	—
pedalion-b-pos	Extra-scheme POS used as a local extension (b)	morphology	S	M	—	8,555	—	1,405	—	2,131
personal-pronoun-person	Person not annotated on a personal pronoun	morphology	S	M	2,372	176	1,773	1	1,451	1
underscore-person-number	Underscore in morphological slots 2-8	morphology	S	A	1,364	220	—	—	—	—
lemma-graphic-variant	Competing graphic variants of the same lemma	lemmatisation	S	M	477	180	6	1	7	3
lemma-suspicious-characters	Anomalous characters in the lemma	lemmatisation	S	M	804	17	4	1	2	—
lemma-tool-numbering	Final number in the lemma	lemmatisation	S	M	72,836	311	—	—	—	—
apostrophe-mixed	Elision apostrophes encoded with different characters	text and encoding	S	A	1,655	2,394	—	—	—	—
corpus-not-nfc	Corpus predominantly in NFD	text and encoding	S	A	—	—	1	1	1	1
unicode-not-nfc	Encoding not normalised to NFC	text and encoding	S	A	—	4,007	—	—	—	—
case-letter-l	Locative, an allowed but rare value	morphology	C	M	2	—	—	—	—	—
common-gender-c	Extra-scheme value c in the gender slot	morphology	C	M	—	8,256	5,163	835	4,267	769
dein-participle-disputed	Participial δεῖν to be checked against the passage	morphology	C	M	1	—	—	—	—	—
mediopassive-d	Voice d not provided for	morphology	C	M	2	—	—	—	—	—
nonlexical-z-pos	PoS outside the scheme on a non-lexical record (z)	morphology	C	M	—	—	—	—	—	—
slot-missing-bare	Noun with no morphological features (n--------)	morphology	C	M	7	35	227	1	59	54
tan-anomaly	τάν/τᾶν, a lexically ambiguous form	morphology	C	M	1	—	—	—	—	—
verbal-adjective-g	Local value g for verbal adjectives in -τέος	morphology	C	M	—	47	—	—	—	—
lemma-rare-for-form	Rare lemma for form and POS	lemmatisation	C	M	102	41	—	2	—	—
uninflected-lemma	Nominal lemma without case in all its occurrences	lemmatisation	C	M	—	1	16	—	7	4
aux-finite-verb	Finite verb with the relation AuxY or AuxZ	syntactic structure	C	M	286	3	2	—	4	—
aux-governor-without-child	AuxP or AuxC without a dependent	syntactic structure	C	M	114	126	877	5	970	17
aux-governs-conjuncts	Non-COORD node governing several _CO conjuncts	syntactic structure	C	M	42	126	129	1	187	8
coord-discordant-conjuncts	Different functions within the same coordination	syntactic structure	C	M	33	242	346	—	451	2
relation-bare-convention	Label outside the scheme used systematically by the corpus	syntactic structure	C	M	—	—	2,239	—	1,782	—
sentence-many-roots	Sentence with several roots	syntactic structure	C	M	1,196	844	—	9	—	91
double-accent	Two accents on the same word	text and encoding	C	M	55	24	38	2	9	3

Identifier	Name	Family	Cl	Tr	Gorman	Pedal.	GL auto.	GL man.	Conf. auto.	Conf. man.
agreement-modifier	Possible failure of agreement in the modifier	agreement and government	C	M	1,593	499	1,281	51	929	44
case-object	Object in an unexpected case	agreement and government	C	M	392	141	57	14	27	1
case-subject	Subject in a case unexpected for the context	agreement and government	C	M	629	267	114	6	110	7
rare-relation-for-lemma-and-pos	Rare relation for lemma and POS	agreement and government	C	M	2,765	918	517	24	824	37
Total	77 checks				101,047	29,515	13,168	2,400	11,259	3,188

Table S1.2 The findings by diagnostic class and by stratum, obtained by summing the rows of Table S1.1. The percentages are computed on the total number of findings of the stratum. These are the figures of Tables 5, 6 and 7 of the article and of Figure 1: Table 5 gives the rows of Gorman and Pedalion, Table 6 those of the two automatic strata of GLAUx, and Table 7 the findings and the verifiable violations of all six strata.

Stratum	Verifiable violation	%	Systematic variant	%	Case to check	%	Total
Gorman	1,914	1.9	91,913	91.0	7,220	7.1	101,047
Pedalion	1,361	4.6	16,584	56.2	11,570	39.2	29,515
GLAUx automatic	378	2.9	1,784	13.5	11,006	83.6	13,168
GLAUx manual	41	1.7	1,409	58.7	950	39.6	2,400
GLAUx confirmation automatic	172	1.5	1,461	13.0	9,626	85.5	11,259
GLAUx confirmation manual	15	0.5	2,136	67.0	1,037	32.5	3,188

Table S1.3 Distribution of the 77 checks by combination of treebank and diagnostic class. The checks are counted as common when they fire in Gorman, in Pedalion and in at least one of the strata of GLAUx. This indicates that the check finds the same type of phenomenon in more than one resource, but it does not prove that it is valid in all treebanks.

Where it fires	Violation	Variant	Check	Total
Gorman, Pedalion and GLAUx	6	3	12	21
only Gorman and/or Pedalion	33	6	5	44
GLAUx and only one of the two	3	1	2	6
only GLAUx	—	1	1	2
no corpus	3	—	1	4
Total	45	11	21	77

By diagnostic class the catalogue comprises 45 verifiable violations, 11 systematic variants and 21 cases to check; by treatment, 19 checks with automatic correction and 58 with manual review (Table 3 of the article); by family, 31 checks of morphology, 20 of syntactic structure, 16 of text and encoding, 6 of lemmatisation and 4 of agreement and government (Table S1.1). No case to check has an automatic correction: where the context decides, there is no single answer to write. Four checks give no finding in any of the six strata. Three of them — beta-code-form, invisible-char-form and invisible-char-lemma — find no occurrence among the active tokens (beta-code-form has a single occurrence in Pedalion, on a technical token) and remain in the catalogue because they describe anomalies that are possible in other resources (§ 4.5 of the article). The fourth, nonlexical-z-pos, examines tokens whose morphological tag begins with z; since such a tag is one of the five criteria that make a token technical (S11.1), with the filter adopted in the study the check never receives a token to examine, and it becomes operative only in a run in which technical tokens are kept.

The eleven systematic variants follow the definition of § 4.1 of the article: a convention different from the reference one but regular within the corpus. They concern how a value, a sign or a headword is written, not whether the linguistic analysis is right, and for this reason they are not verifiable violations; and they are regular practices of the resource, not accidents of a single token that require a reading of the context, and for this reason they are not cases to check. Table S1.4 gives, for each of them, what the resource writes and the reference with which GCP compares it.

The findings fall on 98,440 distinct tokens in Gorman and on 27,709 in Pedalion, that is, on 16.0% and 10.1% of the active tokens, and 2,474 and 1,684 of those tokens carry findings of more than one check (§ 5.1 of the article). The tokens touched include punctuation: in Gorman 62,296 of them are punctuation marks, most of them because of the lemma punc1 (S7.1), and the words touched are 36,144, that is, 6.6% of the 548,912 words; in Pedalion the words touched are 25,312 (10.6%) and the punctuation marks 2,397. The rates per 10,000 active tokens used in the article are therefore densities of findings, not percentages of wrong tokens.

Table S1.4 The eleven systematic variants: what the resource writes and the reference with which GCP compares it.

Check	What the resource writes	Reference adopted by GCP
degree-underscore	the underscore in the ninth position of the morphological tag	the hyphen of the AGDT 2.0 tag
underscore-person-number	the underscore in positions 2–8 of the morphological tag	the hyphen of the AGDT 2.0 tag
grade-p	p (positive) in the position of degree	the hyphen: AGDT 2.0 writes only c and s there
pedalion-b-pos	the part of speech b, declared by the README of the Pedalion Trees	the parts of speech of the AGDT 2.0 inventory
personal-pronoun-person	a personal pronoun without person	the profile of GCP, which fills person on personal pronouns
lemma-tool-numbering	a number appended to the lemma, as in λέγω3 or punc1	the lemma without the number (Celano 2014, § 2.5)
lemma-suspicious-characters	hyphens, brackets or digits inside the lemma	a citation form made only of letters
lemma-graphic-variant	two spellings of the same headword in the same resource	one canonical spelling (list varianti-canoniche-di-lemma, S6)
apostrophe-mixed	two code points for the elision apostrophe in the same file	a single code point
unicode-not-nfc	the form or the lemma not in NFC in fewer than half of the active tokens	NFC
corpus-not-nfc	the form or the lemma not in NFC in at least half of the active tokens (in GLAUx almost always because it is in NFD)	NFC

S2. The specification of the 77 checks

Every check is specified by the five fields named in § 4 of the article: the data it examines, the condition that makes it fire, the numerical parameters it uses, its exclusions and its outcome. The specification, not the code, defines the check: another program that applies these 77 specifications, with the closed lists of S6 and the fixed limits of Table S4.2, to the same files must find the same findings. The rules that hold for all the checks are stated first; the 77 entries follow, in the order of Table S1.1, each with its five fields on separate lines.

Data examined. Each check receives one of the sets of tokens defined in S11.1. The checks of morphology, of lemmatisation and of text and encoding receive the active tokens — the imported tokens minus the technical ones — and most checks of morphology only those whose morphological tag has nine characters. The checks of syntactic structure and of agreement and

government receive the imported tokens sentence by sentence, technical tokens included, because a technical token can be the head of an arc or the root of a sentence; the one exception is the rare relation for lemma and part of speech, which receives the words. The line “Data examined” of each entry names the set and the fields of the token that the check reads to decide whether to report it and what to propose. Two things are not repeated in the entries: the fields that identify technical tokens (S11.1) and those on which a verified repair is recognised (see Outcome).

Exclusions. With the filter adopted in the study (S11.1), no finding falls on a technical token: the checks that receive the active tokens do not see them, and the checks that receive the imported tokens use them in the computation and then discard every finding that falls on a token excluded by the filter. The line “Exclusions” of each entry lists only what that check excludes in addition to this; “None” means that only this common rule applies.

Outcome. The outcome gives the diagnostic class — verifiable violation, systematic variant or case to check — and the treatment: “with automatic correction” when the check provides a single replacement, “for manual review” when it provides none. The treatment is declared for the check; whether an individual finding carries a proposed value can be seen on the finding, and the two can differ in three cases. corpus-not-nfc has automatic correction, but its single finding carries no proposed value, because the conversion into NFC applies to the whole corpus and not to one token. breathing-impossible proposes a value only for the form ὃ with the lemma δέ, which covers all its findings in the six strata. Conversely, some findings of checks for manual review carry a proposal: for 38 occurrences of Gorman, under seven checks, and for 29 occurrences of isolated-breathing in Pedalion, the reading has been established by hand, one occurrence at a time, and the program shows it as a proposal on that occurrence only, without ever applying it automatically. These verified repairs are listed in the program (rule_catalog.py).

One finding per token. Unless the entry says otherwise, a check emits one finding for each token that meets its condition, and a token that meets the conditions of several checks receives one finding from each — up to five in Gorman and in Pedalion. Three checks emit one finding for a group, placed on a representative token: sentence-no-root, one per sentence, on its first token; uninflected-lemma, one per lemma, on the first of its occurrences; corpus-not-nfc, one per run, on the first active token not in NFC. The checks on syntactic nodes report the governing node: coord-discordant-conjuncts the COORD node; aux-governs-conjuncts, aux-governor-without-child and aux-finite-verb the Aux node. Ten checks decide on counts made over the whole run — relation-truncated-or-foreign, relation-bare-convention, corpus-not-nfc, unicode-not-nfc, lemma-graphic-variant, lemma-rare-for-form, uninflected-lemma, slot-missing, slot-missing-bare and rare-relation-for-lemma-and-pos — and apostrophe-mixed on the characters used in each file; except for corpus-not-nfc and uninflected-lemma, they still report token by token.

Two notions of syntax. The base relation is the relation label without the suffixes _CO and _AP and without the infix _ExD with the label that follows it: OBJ_CO, ADV_AP_CO and SBJ_ExD0_ADV count as OBJ, ADV and SBJ. The effective head of a token is the first node reached by climbing from its head past the nodes whose base relation is COORD, APOS, AuxP or AuxC, for at most twelve steps (Table S4.2): a conjunct is thus compared with what the coordination depends on, and the dependent of a preposition with what the preposition depends on; an entry that climbs past a different set of nodes says so. The head is always looked for within the sentence (S11.1): a head of 0 marks the root; an empty head, or one that names no token of the

sentence, has no head token; when several tokens of the sentence share the identifier that a head names, the head is the first of them in the file.

Terminology. As in § 3 of the article, the nine-position string of the morphological analysis is called the morphological tag, and "label" is reserved for the syntactic relation. The identifiers and names of the checks and the names of modules and columns are reproduced as the program writes them (for example missing-postag, postag.py, suggested_postag), even where a name calls the morphological tag "label" or "PoS tag" and the token "record". Grammatical references are to Smyth (1920), by section (§); lexical references are to the lexicon of Liddell, Scott and Jones (1996), henceforth LSJ, by entry (s.v.).

The specification has been checked against the data of the six strata: no finding falls on a technical token, every finding refers to an existing token, and the findings of each check fall within the data declared for it.

Table S2.1 shows how the five fields describe a check, with the two examples of the article: δεῖν annotated as a participle (§ 5.2), a phenomenon covered by two twin checks, and the rare relation (§ 4.3), a single check with two numerical parameters.

Table S2.1 Two examples of how the five fields describe a check. Left, δεῖν annotated as a participle (§ 5.2 of the article): in Gorman δεῖν occurs 165 times among the words, and 37 of these occurrences are annotated as a participle; 36 are reported by dein-participle, which proposes the infinitive automatically, and one, in Lysias 14, by its twin dein-participle-disputed, which sends it to manual review because LSJ (s.v. δεῖ, IV) marks a participial δεῖν in that speech as doubtful (S7.9). Right, the rare relation for a lemma and a part of speech (§ 4.3 of the article).

	dein-participle	dein-participle-disputed	rare-relation-for-lemma-and-pos
Data examined	active tokens with a nine-character morphological tag; form, tag and provenance of the token	the same	words; lemma, part of speech and relation
Condition	the form δεῖν carries p (participle) in the position of mood	the same, in Lysias 14 or in Aristophanes, the works of the two passages that LSJ marks as doubtful, or when the provenance fields of the token give neither a CTS citation nor both the author and the work numbers	a pair of lemma and part of speech occurs at least 200 times, and one of its relations accounts for at most 1% of those occurrences
Parameters	none	none	200 occurrences and 1% (S4)
Exclusions	the occurrences that go to the twin check	the occurrences that go to dein-participle	words with an empty lemma, part of speech or relation
Outcome	verifiable violation, with automatic correction to the infinitive (v--pna---)	case to check, for manual review	case to check, for manual review
Findings	36 in Gorman	1 in Gorman (Lysias 14)	2,765 in Gorman, 918 in Pedalion

creon-anomaly — Nominal χρεὼν with an incomplete PoS

Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The form, compared in NFC, is χρεὼν and the morphological tag is exactly n-_---n_-: a neuter noun with the underscore, which is not a value of the inventory, in the positions of number and case. The program records the reading n-s---nn- (neuter nominative singular) in the parameters of the finding, without proposing it; the reading becomes a proposal, never applied automatically, only on the occurrence of Gorman validated in the table of verified repairs. Any other occurrence must be analysed in context, where a verbal analysis or another reading may be the right one.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

dein-participle — δεῖν with participial mood
Data examined. Active tokens with a nine-character morphological tag. Fields read: form, morphological tag and provenance fields of the token.

Condition. The form δεῖν is annotated with p in the mood position, that is, as a participle. In uncontroversial occurrences, δεῖν is the infinitive of δεῖ and requires n; the program proposes the tag v--pna---.

Parameters. None.

Exclusions. The occurrences that may belong to the two passages that LSJ, s.v. δεῖ, IV, marks as doubtful, which do not leave the catalogue but pass to the twin check dein-participle-disputed. Since the resources carry no reference finer than the work, the list loci-dein-participiale (S6) identifies Lys. 14.7 by the work, so that every participial δεῖν of Lysias 14 (tlg0540.tlg014) passes to the twin; Ar. Fr. 220 cannot be identified, so that every participial δεῖν of Aristophanes (tlg0019) passes to it; and so does every occurrence whose provenance fields give neither a CTS citation nor both the author and the work numbers. The pair therefore covers the phenomenon in full: of the 37 occurrences of δεῖν with p in the mood position attested in Gorman, 36 fall under this check and 1 under its twin, and none is left out (S7.9).

Outcome. Verifiable violation, with automatic correction.

deina-anomaly — Pronominal δεῖν’ with an incomplete PoS
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The form, compared in NFC, is δεῖν’, elided from the pronoun δεῖνα “so-and-so”, and the morphological tag is exactly p-_---__-: a pronoun with the underscore in the positions of number, gender and case. The program records the reading p-s---ma- (masculine accusative singular, as in τὸν δεῖν’) in the parameters of the finding, without proposing it; the reading becomes a proposal, never applied automatically, only on the two occurrences of Gorman validated in the table of verified repairs. Any other occurrence must be analysed in context.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

invalid-character — Character not allowed in the morphological slot
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. A position of the morphological tag contains a character that the AGDT 2.0 inventory does not provide for that position (Table 1 of the article); the hyphen, which marks a feature not filled, is accepted in every position except the first, because every word has a part of speech. The values for which the catalogue has a dedicated check are left to that check and are not reported here: b and z in the first position (pedalion-b-pos, nonlexical-z-pos); the underscore in positions 2–8 (underscore-person-number) and in the ninth (degree-underscore); d in the position of voice (mediopassive-d); c in the position of gender (common-gender-c); l in the position of case (case-letter-l); p in the position of degree (grade-p); in the first position, x on a form that is not a placeholder (leading-x-lexical; the program does not accept x on a placeholder either, but placeholders are technical tokens and do not reach the check), t when the tag carries a value of tense, mood or voice (leading-t) and e on a form made only of punctuation (punctuation-leading-e). Every other character remains with this check, including the hyphen or the underscore in the first position and g in the position of mood: the tokens reported by participle-mood-g and by verbal-adjective-g therefore also receive a finding of this check. In Gorman the 36 findings concern the first position in 26 cases (24 hyphens, one underscore and one Greek δ) and the position of mood in 10 (g); in Pedalion the 62 findings concern the first position in 13 cases (hyphens) and the position of mood in 49 (g 47 times, a and f once each). The check can emit one finding for each offending position; in the six strata no token has more than one. The correct value depends on the morphological analysis of the form and must be checked in context.

Parameters. None.

Exclusions. The values left to the dedicated checks listed above.

Outcome. Verifiable violation, for manual review.

invalid-length-broken — Label of incorrect length to be reconstructed in context
Data examined. Active tokens with a morphological tag of any length other than nine characters; a tag that is absent, empty or made only of spaces is examined by missing-postag instead. Field read: morphological tag.

Condition. The morphological tag does not have the nine characters required and cannot be brought to nine by the mechanical operations of invalid-length-padding, that is, by adding a final hyphen to a tag of eight characters whose positions after the first are all hyphens, or by removing the characters beyond the ninth when they are all hyphens. The missing or extra character may alter the interpretation of the following positions; the whole morphological tag must therefore be reconstructed in context.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

invalid-length-padding — Label consisting only of hyphens with an incorrect length
Data examined. Active tokens with a morphological tag of any length other than nine characters; a tag that is absent, empty or made only of spaces is examined by missing-postag instead. Field read: morphological tag.

Condition. The morphological tag does not have the nine characters required but can be brought to nine by adding or removing hyphens only: either it has eight characters and all the positions after the first are hyphens, and a final hyphen is added; or it is longer than nine and all the characters beyond the ninth are hyphens, and they are removed. The first position is not tested: in Gorman the 320 findings include 62 tags made of eight hyphens and tags of fourteen characters whose first nine positions carry values. A tag that is already nine characters long is not modified. In the proposed tag every underscore is also written as a hyphen: in Gorman four tags of fourteen characters with the underscore in the ninth position, such as a-s---mn_-----, receive the proposal a-s---mn-.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

leading-t — PoS not allowed in slot one (t)
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The first position of the morphological tag contains t, a value not provided for by the AGDT 2.0 morphological inventory, and at least one of the positions of tense, mood and voice carries a value of the inventory; a t on a tag without such values is left to invalid-character. The correct part of speech must be determined on the basis of the form and the context. For the six occurrences of Gorman, validated one by one in the table of verified repairs, the finding shows as a proposal the same tag with v in the first position; the proposal is not marked as automatically correctable.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

leading-x-lexical — PoS x on a lexical record
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The first position of the morphological tag contains x, but the form is a word of the text: it is neither a placeholder — G, G?, G_L or [0] to [20] (Table S4.2) — nor a sequence of combining signs alone. The correct part of speech must be determined in context. For the eighteen occurrences of Gorman, validated one by one in the table of verified repairs, the finding shows a proposed tag, which is not marked as automatically correctable.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

missing-postag — PoS tag missing
Data examined. Active tokens without a morphological tag, whose form is not a placeholder. Fields read: form and morphological tag.

Condition. In the XML file the token has no postag attribute, or the attribute is empty (postag="") or contains only spaces. The morphological tag must be reconstructed from the form, the lemma and the context.

Parameters. None.

Exclusions. The placeholder forms G, G?, G_L and [0] to [20] (Table S4.2).

Outcome. Verifiable violation, for manual review.

numeral-generic-m — Numeral with underscores in the PoS tag
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The morphological tag is exactly m-_---__-: the part of speech is the numeral and the positions of number, gender and case carry the underscore, which is not a value of the AGDT 2.0 inventory. The program replaces the tag with m--------, the tag of a numeral without number, gender and case. The form and the lemma are not examined, and the list of indeclinable numerals (S6) is not used; in the six strata the 28 tokens concerned, 24 in Gorman and 4 in Pedalion, are numerals written with letters or indeclinable cardinals (Smyth § 350). The same underscores are also reported, as a convention of the resource, by underscore-person-number, which proposes the same tag (S3).

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

osfranthiti-anomaly — ὀσφράνθητι with an incompatible morphological analysis
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The form, compared in NFC, is ὀσφράνθητι and its morphological tag is not v2samp---, the tag of the second person singular of the aorist passive imperative, which the program proposes. The correction concerns the morphological tag alone and neither modifies nor certifies the lemma, the syntactic relation or the head.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

participle-mood-g — Value not allowed in the mood slot (g)
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The tag has v in the first position and g in the fifth, a value not provided for verbal mood in the AGDT 2.0 inventory, and the positions of gender and case carry a value (m, f, n or c; n, g, d, a, v or l). The forms recognised as verbal adjectives in -τέος are left to verbal-adjective-g, and the forms covered by a lexical repair to the corresponding check (osfranthiti-anomaly). The tokens reported here also receive a finding of invalid-character (8 in Gorman). The correct value depends on the analysis of the form. The rule derives no replacement tag; the eight occurrences of Gorman, checked by hand one by one in the table of verified repairs, carry in the finding the participial tag established for each of them (for example v-sppamn- for ἔχων), which is not marked as automatically correctable.

Parameters. None.

Exclusions. The verbal adjectives and the lexical repairs just mentioned.

Outcome. Verifiable violation, for manual review.

pot-anomaly — Elided forms πότ’/πόθ’ with an adjectival PoS
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The form, compared in NFC, is πότ’ or πόθ’, written with U+02BC, and the morphological tag begins with a; in the two occurrences of Gorman it is a-_---___. The two forms are elisions of the interrogative adverb πότε “when?”; in πόθ’ the τ becomes θ before a rough breathing. The program records the reading d-------- (adverb) in the parameters of the finding, without proposing it; the reading becomes a proposal, never applied automatically, only on the two occurrences of Gorman validated in the table of verified repairs. Any other occurrence must be analysed in context.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

punctuation-leading-e — PoS e on a punctuation mark
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. Every character of the form, in NFC, belongs to the Unicode punctuation categories (P), but the first position of the morphological tag contains e. For punctuation the encoding adopted uses u; the program therefore proposes the same tag with u in the first position.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

san-anomaly — σὰν as a letter name with an incomplete PoS
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The form, compared in NFC, is σὰν and the morphological tag is exactly n-_---n_-: a neuter noun with the underscore in the positions of number and case. The word is σάν, τό, the Doric name of sigma (LSJ s.v. Σ σ ς, citing Hdt. 1.139). The program records the reading n-s---na- (neuter accusative singular) in the parameters of the finding, without proposing it; the reading becomes a proposal, never applied automatically, only on the occurrence of Gorman validated in the table of verified repairs. Any other occurrence must be analysed in context.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

slot-forbidden — Feature not allowed for the part of speech
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The morphological tag fills a position that its part of speech does not admit, for example a tense on a noun or a case on a verb in a finite mood. The positions that each part of speech requires and admits are those given under slot-missing. A position counts as filled only when it carries one of the values that the program accepts there: 1, 2 or 3 for person; s, p or d for number; p, i, r, l, t, f or a for tense; i, s, o, m, n or p for mood; a, p, m or e for voice; m, f or n for gender; n, g, d, a or v for case; c or s for degree. The underscore and the values that have a check of their own — c in the position of gender, l in that of case, p in that of degree and d in that of voice — do not count, and neither do the characters outside the inventory, which are reported by invalid-character. The feature or the part of speech must be checked in context.

Parameters. None.

Exclusions. The tags whose first position has no profile in the program (slot-missing).

Outcome. Verifiable violation, for manual review.

slot-missing — Required morphological feature left unfilled
Data examined. Active tokens with a nine-character morphological tag. Fields read: form, lemma and morphological tag.

Condition. The morphological tag leaves with the hyphen at least one position that its part of speech requires, for example the gender of a noun or the person of a finite verb, and carries something other than the hyphen in at least one position after the first; a tag with the hyphen in every position after the first passes to slot-missing-bare. A position counts as unfilled only when it carries the hyphen: an underscore is left to underscore-person-number. The analysis is incomplete and must be completed in context. The positions required and admitted depend on the first position and, for the verb, on the mood: the noun and the article require number, gender and case and admit nothing else; the adjective requires the same three and also admits degree; the pronoun requires the same three and also admits person; the numeral requires nothing and admits number, gender, case and degree; the adverb requires nothing and admits only degree; the particle, the conjunction, the preposition, the interjection and the punctuation mark admit no position; x requires nothing and admits every position; the verb in the indicative, subjunctive, optative or imperative requires person, number, tense, mood and voice and admits nothing else; the infinitive requires tense, mood and voice and admits nothing else; the participle, and the verb with d in the position of mood, requires number, tense, mood, voice, gender and case and also admits degree; a verb whose position of mood carries the hyphen or any other value requires nothing and admits every position. Tags whose first position has no profile in the program — in the six strata b, z, t, e, the hyphen, the underscore and, once in Gorman, a Greek δ — are not examined.

Parameters. None.

Exclusions. The indeclinable numerals of the list declared in S6 and the numerals written with Greek letters and the labels of diagrams, recognised by form or by lemma, compared in NFC: a string of Greek letters or numeral signs (such as ʹ, ∠ or the overline) without accents, breathings, iota subscript or diaeresis that ends with a keraia, or with a sign used in its place, or consists of a single character, or whose letters, at most five, are all capitals (Table S4.2); the occurrences of the lemmas that the corpus never inflects, which pass to uninflected-lemma, whose parameter sets that threshold (S4).

Outcome. Verifiable violation, for manual review.

tessereskaideka-anomaly — τεσσερεσκαίδεκα with a non-numeral PoS
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The form, compared in NFC, is τεσσερεσκαίδεκα or τεσσαρεσκαίδεκα, the indeclinable numeral “fourteen”, and the morphological tag does not begin with m, the numeral. The program proposes m--------.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

missing-lemma — Lemma missing
Data examined. Active tokens. Field read: lemma.

Condition. The lemma field is missing, empty or made only of spaces. The correct lemma must be established from the form and the context and cannot be reconstructed safely by a mechanical substitution.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

apos-suffix-orphan — The suffix _AP without an apposition
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head and relation.

Condition. The relation ends in _AP: the label declares that this word and another expression denote the same person or thing, as in “Socrates, the philosopher”. In the annotation style that uses the suffix, the label APOS marks the node that separates the noun from its appositive, and the two members carry the function required in the sentence with the suffix _AP (Celano 2014, § 3.10). The program climbs from the head of the token past the nodes whose base relation is COORD, AuxP or AuxC and examines the first node that is not one of these. If the relation of that node neither begins with APOS nor ends in _AP or _CO — the last two cases cover nested appositions and coordinations — the token is reported: the suffix declares an apposition that the tree does not represent, and the suffix or the links must be corrected

by hand. Labels in _AP_CO, which mark coordinated appositives (Celano 2014, § 3.20.2), end in _CO and are examined by coord-suffix-orphan.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

coord-suffix-orphan — The suffix _CO without a coordinating node
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head and relation.

Condition. The relation ends in _CO, the suffix that marks the members of a coordination, whose node carries the relation COORD (Celano 2014, § 3.9; Bamman and Crane 2008: 26–27). The program climbs from the head of the token past the nodes whose base relation is AuxP, AuxC, AuxY or APOS and examines the first node that is not one of these. AuxP and AuxC are passed because, when prepositional phrases or subordinate clauses are coordinated, the suffix stands on the child of the preposition or of the conjunction; AuxY because the correlative conjunctions that precede the coordinating one are attached to it with that label. If the relation of the node reached neither begins with COORD nor ends in _CO or _AP — the last two cases cover nested coordinations and appositions — the token is reported: the suffix declares a coordination that the tree does not represent, and the suffix or the links must be corrected by hand.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

head-cycle — Cycle in the heads
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier and head.

Condition. The token lies on a cycle of heads: following the head values from the token, the program comes back to the token itself without reaching the root, an empty head or a head that names no token of the sentence. The cycle may consist of a single token that indicates itself as its head or of several tokens that depend on one another in a circle; in either case the structure is not a valid tree. Each token of the cycle receives a finding, whereas the tokens whose chain of heads leads into a cycle without belonging to it are not reported.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

head-dangling — Non-existent head
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier and head.

Condition. The head field of the token contains a number other than 0. The program looks in the same sentence for a token whose identifier is that number; if it does not find one, the syntactic link points to a non-existent head and is not valid. The tree must therefore be reconstructed manually.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

head-missing — Syntactic head not declared
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head and relation.

Condition. The head field of the token is empty (the identifier of the word on which the token depends is therefore not given), while at least one other token in the sentence, technical tokens included, declares a head. Tokens whose relation

is AuxG, AuxK or AuxX, written without suffixes, are not examined, as in sentence-not-annotated. The correct head must be indicated manually.

Parameters. None.

Exclusions. The tokens whose relation is AuxG, AuxK or AuxX.

Outcome. Verifiable violation, for manual review.

relation-greek-lookalike — Look-alike Greek character in the syntactic label
Data examined. Imported tokens, sentence by sentence, technical tokens included. Field read: relation.

Condition. The relation field contains a Greek character graphically identical to the expected Latin character, for example the Greek capital zeta Z (U+0396) in place of the Latin Z of AuxZ. The program replaces the character with its Latin counterpart, provided that the label so obtained is a label of the scheme.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

relation-missing — Relation missing
Data examined. Imported tokens, sentence by sentence, technical tokens included. Field read: relation.

Condition. The relation field is missing, empty or made only of spaces. The syntactic function of the token must be determined by checking the sentence and the tree.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

relation-nil-placeholder — The placeholder nil in place of the relation
Data examined. Imported tokens, sentence by sentence, technical tokens included. Field read: relation.

Condition. The relation field contains nil, a placeholder that indicates the absence of the analysis and not an AGDT syntactic relation. The correct relation must be established by checking the sentence and the tree.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

relation-truncated-or-foreign — Non-AGDT relation traceable to a known convention
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: relation and document.

Condition. The relation field, stripped of surrounding spaces and with every Greek letter that the program treats as a look-alike of a Latin letter replaced by that letter, is exactly one of three labels that AGDT 2.0 does not recognise but that can be traced to a known convention: CO, a conjunct suffix left without the function that should precede it; Aux, a truncated auxiliary label; CONJ, a label taken from another scheme. The comparison concerns the whole field and is case-sensitive: AUX, which occurs 26 times in Pedalion, and labels with a suffix such as Aux_CO are not recognised and are reported by relation-unknown. CO on its own is not admitted: in the AGDT scheme _CO is a suffix appended to the function of each conjunct (Celano 2014, § 3.20.1), as in the analytical layer of the Prague Dependency Treebank (Hajič et al. 1999). The program proposes no automatic correction: the correct relation must be chosen by reading the sentence and checking the tree.

Parameters. The label, counted as it is written, occurs fewer than 20 times among the imported tokens of the run, technical tokens included, or in less than 90% of the documents of the run (BARE_RELATION_MINIMUM and BARE_RELATION_SPREAD, S4).

Exclusions. The labels that reach both thresholds, which pass to relation-bare-convention.

Outcome. Verifiable violation, for manual review.

relation-typo — Unambiguous typo in the syntactic label
Data examined. Imported tokens, sentence by sentence, technical tokens included. Field read: relation.

Condition. The relation field, stripped of surrounding spaces and with every Greek letter that the program treats as a look-alike of a Latin letter replaced by that letter, is one of ten misspellings, each with a single intended label: AucZ and AuxZZ for AuxZ, AucY for AuxY, Coord and coord for COORD, Pred for PRED, Sbj for SBJ, Obj for OBJ, Atr for ATR and Adv for ADV. The program replaces the value with the intended label. The list is closed: labels that differ from a label of the scheme by a single character but are not in it, such as MWE2 and fAuxY, are reported by relation-unknown. In the six strata the check fires only once, on AucZ in Gorman.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

relation-unknown — Relation outside the scheme
Data examined. Imported tokens, sentence by sentence, technical tokens included. Field read: relation.

Condition. The relation field is filled and its base relation is not one of the 23 labels that the program accepts: PRED, SBJ, OBJ, ATR, ADV, PNOM, OCOMP, ATV, AtvV, COORD, APOS, ExD, MWE, AuxP, AuxC, AuxR, AuxV, AuxX, AuxG, AuxK, AuxY, AuxZ and UNDEFINED. The labels recognised by another check are left to it: nil and NIL (relation-nil-placeholder); CO, Aux and CONJ (relation-truncated-or-foreign or relation-bare-convention); the ten misspellings of relation-typo; the labels that become a label of the scheme once their Greek look-alike letters are replaced (relation-greek-lookalike); and the working notes of annotator-note-in-relation (Table S4.2). In the six strata what remains are labels such as AUX, PUNCT, GAP, MWE2, Aux_CO, PRED_C0, Auxy, AuxZAuxZ and fAuxY. The correct relation must be chosen in context.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

sentence-no-root — Sentence without a root
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head and relation.

Condition. No token of the sentence is a root, a root being a token with head="0" whose relation is not AuxG, AuxK or AuxX: the structural punctuation attached to the root does not count as a root, whereas a technical token does. The structure therefore has no declared root and may contain a cycle, a missing head or a damaged link. The finding is one per sentence and is placed on its first token; when that token is technical, the finding is discarded with it, which happens for one sentence of Pedalion, eight of the automatic stratum of the first GLAUx sample and eighteen of that of the confirmation sample.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

sentence-not-annotated — No head declared in the sentence
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head and relation.

Condition. No token of the sentence has a declared head, so the sentence contains no syntactic link. The program emits one finding for every token without a head whose relation is not AuxG, AuxK or AuxX; in the six strata the three sentences concerned, all in Pedalion, consist of a single token each. The tree must be reconstructed or recovered manually.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

token-id-duplicate — Duplicate identifier
Data examined. Imported tokens, sentence by sentence, technical tokens included. Field read: identifier.

Condition. Two or more tokens in the same sentence have the same identifier. Every reference to that value becomes ambiguous, and the identifiers or the links must be corrected manually.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

annotator-note-in-relation — Annotator's note in the relation field
Data examined. Active tokens. Field read: relation.

Condition. The relation field contains a working note, recognised by one of two signs wherever it occurs in the field: a run of two or more of the characters !, ? and %, as in !!!!!!ATR or %%%SBJ, or four or more lower-case Latin letters followed by a space, as in !!!congruentiefout geslacht?!!!ADV (Table S4.2). The note may be accompanied by a syntactic label, but the value is not a valid relation and the note may contradict the label given; the correct relation must be established by checking the sentence and the tree.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

beta-code-form — Beta code residues in the form
Data examined. Active tokens. Field read: form.

Condition. The form contains at least one of the characters with which beta code writes capitals, breathings, accents, iota subscript and diaeresis — the asterisk, the round brackets, the slash, the backslash, the vertical bar, the plus sign and the equals sign — together with at least one Latin or Greek letter. A value without letters, such as a bracket or a slash tokenised on its own, or a placeholder [n], is not examined. The conversion into Unicode may be incomplete or ambiguous and must be checked manually.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

beta-code-lemma — Beta code residues in the lemma
Data examined. Active tokens. Field read: lemma.

Condition. The lemma contains at least one of the characters with which beta code writes capitals, breathings, accents, iota subscript and diaeresis — the asterisk, the round brackets, the slash, the backslash, the vertical bar, the plus sign and the equals sign — together with at least one Latin or Greek letter. A value without letters, or a placeholder [n], is not examined. The conversion into Unicode may be incomplete or ambiguous and must be checked manually.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

breathing-impossible — Elision apostrophe encoded as a breathing
Data examined. Active tokens. Fields read: form and lemma.

Condition. The form, decomposed, ends with the smooth breathing (U+0313) written on a Greek consonant other than rho, as in δ̓ for δ'. Breathings are written only on initial vowels and diphthongs and on ρ (Smyth §§ 9, 13), and the place of an elided vowel is marked by the apostrophe (Smyth § 70): the breathing stands here for the apostrophe. A form that consists of the sign alone is not examined. The program proposes the apostrophe only for the form δ̓ with the lemma δέ,

whose reading is unambiguous: this covers all 19 findings of Gorman. Any other form would be reported without a proposal, for reading in context.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

final-sigma-inside — Final sigma inside the word
Data examined. Active tokens. Field read: form.

Condition. The character ς appears in the form in a position other than the last, once any elision apostrophes and the signs . , ; · : at the end of the form have been set aside; in that position σ is expected, and the program replaces the final sigma with the non-final one. Forms shorter than two characters, forms whose remaining core has at most two characters (Table S4.2) and forms that contain a letter outside the Greek blocks, such as the G with which Pedalion marks a lacuna in GGαGρωςGG, are not examined.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

invisible-char-form — Format characters in the form
Data examined. Active tokens. Field read: form.

Condition. The form field contains one of the Unicode format characters recognised by the program (U+200B, U+200C, U+200D, U+2060 or U+FEFF). These characters have no orthographic function in the representation of Greek adopted by the corpus and can break the comparison between graphically equivalent strings. The program proposes their removal. Spaces and other separators are handled by separate checks, and a form that also contains a separator is reported only by whitespace-in-form.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

invisible-char-lemma — Format characters in the lemma
Data examined. Active tokens. Field read: lemma.

Condition. The lemma field contains one of the Unicode format characters recognised by the program (U+200B, U+200C, U+200D, U+2060 or U+FEFF). These characters have no orthographic function in the representation of Greek adopted by the corpus and can break the comparison between graphically equivalent lemmas. The program proposes their removal. Spaces and other separators are handled by separate checks, and a lemma that also contains a separator is reported only by whitespace-in-lemma.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

isolated-breathing — Isolated breathing in the form
Data examined. Active tokens. Field read: form.

Condition. The form, stripped of surrounding spaces, consists only of the combining rough breathing (U+0314) or the combining smooth breathing (U+0313). In Pedalion, where the check finds 31 such forms, 29 have been validated one by one in the table of verified repairs: in each of them the sign stands for a round bracket encoded as a breathing, and the finding shows the bracket as a proposal, never applied automatically. Any other occurrence must be reconstructed by hand in context: the sign may also be a fragment detached from a damaged word.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

mixed-script — Look-alike Latin letters in a Greek word
Data examined. Active tokens. Field read: form.

Condition. The form mixes Greek letters with letters of another script, Latin or Cyrillic, and at least one of these has a graphically identical Greek counterpart in the program's table of look-alike characters, as in the two findings of Pedalion, κακῶν, with the Latin v for ν, and Τροίαν, with the Latin T. The program replaces the look-alike letters with their Greek counterparts, so that the form can be compared exactly with the other forms; when no letter has a look-alike, the form goes to mixed-script-unrepairable.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, with automatic correction.

mixed-script-unrepairable — Latin letters that are not look-alikes in a Greek word
Data examined. Active tokens. Field read: form.

Condition. The form mixes Greek letters with letters of another script, Latin or Cyrillic, and none of these has a Greek counterpart in the program's table of look-alike characters, so that no substitution is possible; when at least one has, the form goes to mixed-script. The foreign letter may be an editorial sign, such as the G with which Pedalion marks a lacuna, or an encoding error; the program reports the form without proposing a substitution.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

whitespace-in-form — Internal separator in the form
Data examined. Active tokens. Field read: form.

Condition. The form field contains a space, a no-break space, a tab or a line break. The token may contain more than one word or a damaged segmentation; simply deleting the separator is not a safe correction, and the tokenisation must be reviewed manually.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

whitespace-in-lemma — Internal separator in the lemma
Data examined. Active tokens. Field read: lemma.

Condition. The lemma field contains one or more internal separators: a space, a no-break space, a tab or a line break (the separators at the two ends are removed on import). A lemma is a single headword; whether the value stands for a multi-word expression, a wrong tokenisation or a convention of the resource must be decided by hand, and the program does not delete the separators.

Parameters. None.

Exclusions. None.

Outcome. Verifiable violation, for manual review.

degree-underscore — Underscore in the degree slot
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The ninth position of the morphological tag contains _ instead of -. The program proposes the tag with every underscore replaced by a hyphen, preventing equivalent tags from being counted as different values; where positions 2–8 also contain underscores (1,316 tokens in Gorman, one in Pedalion), the proposal coincides with that of underscore-person-number.

Parameters. None.

Exclusions. None.

Outcome. Systematic variant, with automatic correction.

grade-p — Positive degree marked with p
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The ninth position of the morphological tag contains p to indicate the positive degree, but the encoding adopted uses the hyphen when the degree is neither comparative nor superlative. The program replaces p with - in the ninth position.

Parameters. None.

Exclusions. None.

Outcome. Systematic variant, with automatic correction.

pedalion-b-pos — Extra-scheme POS used as a local extension (b)
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The first position of the morphological tag contains b, a part of speech that the AGDT 2.0 inventory does not provide for and that the README of the Pedalion Trees declares as its own. The program proposes no conversion.

Parameters. None.

Exclusions. None.

Outcome. Systematic variant, for manual review.

personal-pronoun-person — Person not annotated on a personal pronoun
Data examined. Active tokens with a nine-character morphological tag. Fields read: form, lemma and morphological tag.

Condition. The token is a personal pronoun, but the morphological tag does not specify the person. The token is recognised as a personal pronoun when its tag begins with p- (pronoun without person), its form in NFC coincides exactly, accents and capitalisation included, with one of the 36 forms of the list pronomi-personali (S6), and its lemma, compared in NFC, is not ὅς: οὗ is both the pronoun of the third person and the genitive of the relative pronoun, and the condition on the lemma leaves the relative out (in GLAUx, whose lemmas are in NFD, the comparison in NFC is what excludes 145, 8, 138 and 13 occurrences of οὗ lemmatised ὅς in the four strata). Forms absent from the list, such as the grave-accented σὺ, σοὶ and σὲ or the enclitic μοι, σου and σοι, are not examined: among the tokens with a tag beginning with p-, 747 in Gorman and 225 in Pedalion have as lemma ἐγώ, σύ, ἡμεῖς, ὑμεῖς, σφεῖς or ἕ and a form absent from the list, among them the enclitic οἱ of ἕ, 145 times in Gorman and 88 in Pedalion. AGDT 2.0 does not require person on pronouns (Celano 2014, § 2.3), so its absence is not a violation of the scheme (§ 6.3 of the article). The reference profile of GCP fills it, because it makes the annotation complete and comparable across resources: a corpus that leaves it empty follows a different but regular convention, and the check records a systematic variant (S3). No correction is proposed and the tag is left unchanged; the finding records as a parameter the person of the list in which the form appears.

Parameters. None.

Exclusions. None.

Outcome. Systematic variant, for manual review.

underscore-person-number — Underscore in morphological slots 2-8
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The morphological tag contains _ in one of the positions from the second to the eighth, in which the encoding adopted uses - for a feature that does not apply or is not filled. The program emits one finding per token, whatever the number of positions concerned, and proposes the tag with every underscore replaced by a hyphen, including one in the ninth position; the underscore in the ninth position is reported by degree-underscore, so that a tag such as a-_---___ receives both findings (1,316 tokens in Gorman, one in Pedalion), and both propose a--------. Any remaining morphological inconsistencies must be checked separately.

Parameters. None.

Exclusions. None.

Outcome. Systematic variant, with automatic correction.

lemma-graphic-variant — Competing graphic variants of the same lemma
Data examined. Active tokens with the lemma field filled. Field read: lemma.

Condition. The lemma of the token, normalised to NFC and stripped of surrounding spaces and of any final number, is one of the twelve spellings that the list varianti-canoniche-di-lemma (S6) maps onto a canonical spelling — for example ποτέ onto ποτε or ἀείρω onto αἴρω — and the canonical spelling, compared in NFC, is the lemma of at least one active token of the same run. The rule is triggered only when both spellings coexist in the resource; because the comparison is made in NFC, it also fires in GLAUx, which is distributed in NFD. A lemma of the list that ends with a number is examined here and not by lemma-tool-numbering. The choice of the canonical form is a normalisation decision.

Parameters. None.

Exclusions. None.

Outcome. Systematic variant, for manual review.

lemma-suspicious-characters — Anomalous characters in the lemma
Data examined. Active tokens with the lemma field filled. Fields read: lemma and morphological tag.

Condition. The lemma field contains a hyphen or a dash (U+002D or U+2010–U+2014) or a round, square or curly bracket in any position, the first and the last included, or a digit in any position other than the last: a single final digit is left to lemma-tool-numbering, while the first digit of a final number of two or more digits is reported here; a lemma consisting only of a dash or of digits, such as — and 25 in Pedalion, is also reported. These characters may depend on conventions of the source resource (e.g. περι-κεῖμαι, Ἰων(ν)ης); the canonical form must be chosen according to the lexicographical policy adopted.

Parameters. None.

Exclusions. The tokens with a morphological tag beginning with u, whose lemma may legitimately be a sign.

Outcome. Systematic variant, for manual review.

lemma-tool-numbering — Final number in the lemma
Data examined. Active tokens with the lemma field filled. Field read: lemma.

Condition. The lemma, normalised to NFC and stripped of surrounding spaces, ends with a digit: the number with which the resource distinguishes homographic entries (λέγω1, λέγω3) or marks punctuation (punc1). A lemma that, without the final number, is one of the spellings of the list varianti-canoniche-di-lemma is examined by lemma-graphic-variant instead. The program proposes no change, because removing the number may merge entries that the resource keeps distinct; the removal belongs to the normalisation policy that the user declares.

Parameters. None.

Exclusions. None.

Outcome. Systematic variant, for manual review.

apostrophe-mixed — Elision apostrophes encoded with different characters
Data examined. Active tokens. Fields read: form and source file.

Condition. Within the same source file, the forms use more than one of the nine characters that the program treats as an elision apostrophe: the ASCII apostrophe, U+2019, U+02BC, U+1FBD, U+1FBF, U+00B4, U+02B9, U+2032 and U+1FEF. In such a file, every token whose form contains one of these characters other than ’ (U+02BC) receives a finding, and the program proposes to convert the character into ’, adopted as the canonical character following the normalisation that Celano (2025: 50) applies to his training data. The substitution makes the encoding uniform, but it is not a requirement of the AGDT scheme and is applied only if the user accepts the proposal.

Parameters. None.

Exclusions. The files that use a single character.

Outcome. Systematic variant, with automatic correction.

corpus-not-nfc — Corpus predominantly in NFD
Data examined. All the active tokens of the run. Fields read: form and lemma.

Condition. A token is counted when its form or its lemma, if not empty, differs from its own NFC normalisation. If the tokens counted are at least 50% of the active tokens of the run, the encoding is treated as a property of the resource: the check emits a single finding, on the first active token counted, and gives the count in its parameters. The correction declared by the catalogue is the conversion into NFC of the whole corpus, which does not alter the graphic content: the finding carries no proposed value and is not marked as automatically correctable, because the conversion is not made token by token; when the check is enabled at the correction step, as it is by default for the checks that the catalogue declares automatic, the program converts the form and the lemma of every token of the corpus into NFC. GLAUx, the only resource in which the check fires, declares that its corpus "is encoded in Unicode-NFD" (GLAUx README), which is consistent with the classification as a systematic variant.

Parameters. Share of 50% of the active tokens (WHOLE_CORPUS_ENCODING, S4), reached when the share is equal to or greater than the threshold.

Exclusions. Below the threshold the findings are per token and are produced by unicode-not-nfc.

Outcome. Systematic variant, with automatic correction.

unicode-not-nfc — Encoding not normalised to NFC
Data examined. All the active tokens of the run. Fields read: form and lemma.

Condition. The count is that of corpus-not-nfc: a token is counted when its form or its lemma, if not empty, differs from its own NFC normalisation. When the tokens counted are less than 50% of the active tokens of the run, each of them receives one finding, whichever of the two fields is not in NFC. The program normalises the values concerned, preventing strings that look identical from being counted as different. In Pedalion, the only corpus in which the check fires, most of the 4,007 tokens concerned are not decomposed: 2,195 consist only of the ano teleia U+0387, which NFC replaces with U+00B7, and most of the others contain vowels with oxia (U+1F71–U+1F7D), which NFC replaces with the vowels with tonos; only 18 values, all lemmas, are in NFD.

Parameters. None of its own.

Exclusions. At or above the 50% threshold there is a single finding, produced by corpus-not-nfc.

Outcome. Systematic variant, with automatic correction.

case-letter-l — Locative, an allowed but rare value
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The eighth position of the morphological tag contains l, a value provided for the locative but rare in the corpus. The finding does not imply an error: the occurrence must be checked in context without a predefined correction.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

common-gender-c — Extra-scheme value c in the gender slot
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The seventh position of the morphological tag contains c, a value not provided for gender by the AGDT 2.0 inventory. In the resources analysed the value occurs above all with pronouns and some nouns, and its distribution is compatible with a local convention for the common gender, but this interpretation is not documented in the resources available. The value must therefore be checked against the profile of the individual resource and in context; the program proposes no automatic conversion.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

dein-participle-disputed — Participial δεῖν to be checked against the passage
Data examined. Active tokens with a nine-character morphological tag. Fields read: form, morphological tag and provenance fields of the token.

Condition. The form δεῖν carries a participle tag in a case that the program does not correct without verification: the occurrence belongs to Lysias 14, the work that contains the passage Lys. 14.7, which LSJ, s.v. δεῖ, IV, reports as doubtful; or it belongs to Aristophanes, for whom the doubtful passage Ar. Fr. 220 cannot be identified in the resources; or its provenance fields give neither a CTS citation nor both the author and the work numbers, so that it cannot be excluded that it belongs to one of the two passages. In these cases the program proposes no correction: the analysis must be checked against the passage and the textual tradition. The rule remains separate from the one that corrects the other occurrences of δεῖν annotated as a participle.

Parameters. None.

Exclusions. The occurrences whose provenance excludes the two passages, which pass to the twin check dein-participle. In Gorman a single occurrence reaches this check, in Lysias 14 (S7.9).

Outcome. Case to check, for manual review.

mediopassive-d — Voice d not provided for
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The sixth position of the morphological tag contains d, a value not provided for by the AGDT 2.0 encoding adopted for Greek. It may be an imported or local convention, for example for deponents; the interpretation and the correction must be checked in context.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

nonlexical-z-pos — PoS outside the scheme on a non-lexical record (z)
Data examined. Active tokens with a nine-character morphological tag. Field read: morphological tag.

Condition. The first position of the morphological tag contains z, a value not documented in the AGDT 2.0 inventory. If, in the specific resource and version, the value occurs exclusively on non-lexical tokens, it must be treated as a local technical marker, without automatically inferring a substitute part of speech. With the filter configuration adopted in the study, which counts every token whose tag begins with z as technical (S11.1), the check receives no tag beginning with z and cannot fire: it becomes operative only in a run in which technical tokens are not excluded, where in the six strata it would report 15,608 tokens, all technical. Its zero in Table S1.1 therefore says nothing about the presence of z in the resources.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

slot-missing-bare — Noun with no morphological features (n--------)
Data examined. Active tokens with a nine-character morphological tag. Fields read: form, lemma and morphological tag.

Condition. The morphological tag carries the hyphen in every position after the first, and its part of speech requires at least one of those positions according to the profiles given under slot-missing; tags such as d--------, c--------, m-------- or v-------- are therefore not reported. In the six strata 375 of the 383 findings are nouns written n--------; the other eight are tags a--------: in Gorman five adverbs in -ως lemmatised under the adjective (εὐθέως, κοινῶς, βιαίως, δικαίως, φρονίμως), ὅτι and the numeral δεκαδύο, in Pedalion the particle κε. The tag is formally well formed, but the

morphological analysis is incomplete; the features must be reconstructed in context, except for indeclinable nouns or declared conventions of the resource.

Parameters. None.

Exclusions. The indeclinable numerals of the list declared in S6 and the numerals written with Greek letters, recognised by form or by lemma as in slot-missing; the occurrences of the lemmas that the corpus never inflects, which pass to uninflected-lemma (S4).

Outcome. Case to check, for manual review.

tan-anomaly — τάν/τᾶν, a lexically ambiguous form
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The form, compared in NFC, is τάν, written with the acute accent, and the morphological tag is exactly n-_---__-. The form may belong to the formula of address ὦ τᾶν, the only use that LSJ (s.v. τᾶν) records for the word, or be the Doric accusative singular of the feminine article (Smyth § 332 D); the written form does not decide between the two. The program records the reading i-------- (interjection) in the parameters of the finding, without proposing it, and no occurrence is validated in the table of verified repairs: the analysis must be checked in context. τᾶν with the circumflex, τὰν with the grave and τάν with any other tag are not examined.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

verbal-adjective-g — Local value g for verbal adjectives in -τέος
Data examined. Active tokens with a nine-character morphological tag. Fields read: form and morphological tag.

Condition. The tag has v in the first position and g in the position of mood, and the form, without diacritics, ends in -τε- followed by an adjectival ending (-τεος, -τεου, -τεω, -τεον, -τεοι, -τεων, -τεους, -τεοις, -τεα, -τεας, -τεαι, -τεαν, -τεαις). The resource uses g in the position of mood as a local code for the verbal adjectives in -τέος, which the grammar treats as declined adjectives (Smyth §§ 471–473, 2149–2152) and AGDT 2.0 annotates as adjectives. The program does not propose a replacement tag: every occurrence must be checked in context, because some features cannot be derived from the form alone; in particular, in the neuters in -τέον, such as ποιητέον, the impersonal construction requires the nominative (Smyth § 2152), while in the personal construction the verbal adjective agrees with its subject (§ 2151). The 47 tokens of Pedalion also receive a finding of invalid-character, which does not exclude g in the position of mood.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

lemma-rare-for-form — Rare lemma for form and POS
Data examined. Active tokens with both form and lemma filled. Fields read: form, part of speech and lemma.

Condition. The tokens are grouped by form, as written in the file, and by part of speech, the first character of the morphological tag. In a group of at least twenty-five occurrences, a lemma that occurs once only and differs from the majority lemma of the group is reported when the majority lemma covers at least 95% of the occurrences of the group. The lemmas are compared after NFC normalisation and after the removal of the final number of the lemma, which has its own check (lemma-tool-numbering), but without the list of graphic equivalences of lemma-graphic-variant, which would remove before the comparison the internal differences that this check is meant to observe. The case may be correct or anomalous; the program shows the majority lemma as a possible alternative, but applies no correction automatically.

Parameters. Minimum support 25 occurrences; at most one rare occurrence; minimum majority 95%.

Exclusions. None.

Outcome. Case to check, for manual review.

uninflected-lemma — Nominal lemma without case in all its occurrences
Data examined. Active tokens with a nine-character morphological tag, grouped by lemma. Fields read: form, lemma and morphological tag.

Condition. A lemma, compared in NFC, has at least five occurrences with a nominal part of speech (n, a, m, p or l) among the tokens examined, and none of its nominal occurrences carries anything but the hyphen in the position of case; occurrences with another part of speech do not count towards the five. Its occurrences that slot-missing and slot-missing-bare would report are reported here instead, once per lemma, with the number of occurrences concerned, on the first of those occurrences in the order of the table. This behaviour may depend on the indeclinable or foreign nature of the lemma, on a convention of the resource or on a systematic gap in the annotation, and must be assessed at the level of the lemma.

Parameters. Minimum of five nominal occurrences of the lemma (INDECLINABLE_MINIMUM, S4).

Exclusions. The indeclinable numerals of the list declared in S6 and the numerals written with Greek letters, recognised by form or by lemma as in slot-missing.

Outcome. Case to check, for manual review.

aux-finite-verb — Finite verb with the relation AuxY or AuxZ
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head, relation and morphological tag.

Condition. The morphological tag identifies a verb in a finite mood — v in the first position and, in the fifth, i, s, o or m: indicative, subjunctive, optative or imperative — the relation is AuxY or AuxZ, written without suffixes, and the token governs at least one other token. These functions are not normally assigned to a finite verb in the AGDT 2.0 scheme (Celano 2014, §§ 3.17–3.18), and a finite verb with its own dependents under one of them may reflect a parenthetical or adverbial use; the case must be assessed in context. A finite verb under AuxY or AuxZ that governs no token is not reported.

Parameters. None.

Exclusions. The tokens already reported by aux-governs-conjuncts.

Outcome. Case to check, for manual review.

aux-governor-without-child — AuxP or AuxC without a dependent
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head and relation.

Condition. A token whose relation is AuxP or AuxC, written without suffixes, has no dependents: no token of the sentence, technical tokens included, has it as its head. Normally the complement or the head of the subordinate clause depends on these nodes, but the dependent may be missing because of ellipsis, a lacuna or interrupted text; the case must be checked without an automatic correction.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

aux-governs-conjuncts — Non-COORD node governing several _CO conjuncts
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head and relation.

Condition. A token whose relation is one of the labels of the auxiliary leaf nodes — AuxK, AuxX, AuxG, AuxY or AuxZ, written without suffixes — governs at least one token whose relation ends in _CO; one conjunct is enough, although the name of the check speaks of several. In the AGDT scheme the members of a coordination depend on the node with the relation COORD (Celano 2014, § 3.9): the structure of the tree and the labels therefore give incompatible indications, and the case must be checked in context. The finding falls on the governing node, and its parameters give the number of conjuncts it governs.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

coord-discordant-conjuncts — Different functions within the same coordination
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head and relation.

Condition. Two or more tokens whose relation ends in _CO reach the same coordinating node and have different base relations. As in coord-suffix-orphan, the program climbs from the head of each such token past the nodes whose base relation is AuxP, AuxC, AuxY or APOS, but not past COORD, for at most twelve steps (Table S4.2), and takes the first node that is not one of these; that node is a coordinating node when its relation begins with COORD. A nested coordination that is itself a conjunct, labelled COORD_CO, takes part with the base relation COORD. The finding is emitted on the coordinating node. The asymmetry may be correct or may depend on inconsistent annotation; the program reports the case without proposing a correction.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

relation-bare-convention — Label outside the scheme used systematically by the corpus
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: relation and document.

Condition. One of the three labels recognised by relation-truncated-or-foreign — CO, Aux or CONJ — occurs at least twenty times and in at least 90% of the documents of the run. The distribution suggests a local convention, but does not make it possible to expand the value automatically into an AGDT relation. The thresholds are applied to these three labels only: any other label outside the inventory is reported by relation-unknown however often and however widely it occurs, as MWE2 is in the manual stratum of the first GLAUx sample, where it occurs 21 times, in both documents.

Parameters. At least 20 occurrences among the imported tokens of the run, technical tokens included, and presence in at least 90% of its documents (BARE_RELATION_MINIMUM and BARE_RELATION_SPREAD, S4).

Exclusions. The occurrences of the three labels that do not reach twenty occurrences or 90% of the documents, which pass to relation-truncated-or-foreign.

Outcome. Case to check, for manual review.

sentence-many-roots — Sentence with several roots
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: identifier, head and relation.

Condition. More than one token of the same sentence is a root, in the sense of sentence-no-root: head="0" and a relation other than AuxG, AuxK or AuxX, technical tokens included. An AGDT sentence must have a single root; the case may indicate a structural error or a sentence segmentation to be revised. The check emits a finding on every root of the sentence, not only on the additional ones; the findings that fall on technical roots are then discarded, so that in 82 Gorman sentences only one root keeps its finding.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

double-accent — Two accents on the same word
Data examined. Active tokens. Fields read: the form; the form and the lemma of the next active token of the same sentence; for the exclusion below, the attribute artificial and the relations of the sentence.

Condition. The form, decomposed, carries at least two accents — acute, grave or circumflex — outside the contexts in which a word can receive the accent of an enclitic (Smyth §§ 181–185). The second accent is admitted when the next active token of the same sentence is an enclitic, that is, when its form or its lemma, lower-cased and stripped of diacritics and of any final elision apostrophe, is one of the entries of the list enclitiche (S6) reduced in the same way or,

if it ended with an apostrophe, coincides with an entry ending in ε or α deprived of that vowel, as μ᾽ for με, γ᾽ for γε and τ᾽ for τε. It is also admitted when the form itself, reduced in the same way, ends with one of the enclitics δε, περ, γε, τε, ποτε, που, πω or θι written together with it, as in Ἐλευσῖνάδε or οἷόνπερ, and is longer than that ending by more than two characters (Table S4.2). The case must be checked in context.

Parameters. None.

Exclusions. Elliptic tokens, that is, tokens with the attribute artificial, with a placeholder form from [0] to [20] or placed after the last AuxK of the sentence, where the corpora place the words supplied by the annotator: such a word keeps the accent of its original context, but no enclitic follows it.

Outcome. Case to check, for manual review.

agreement-modifier — Possible failure of agreement in the modifier
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: morphological tag, lemma, relation, head and identifier; for the exclusion of elliptic tokens, also the form and the attribute artificial.

Condition. An article, an adjective or a numeral — l, a or m in the first position of a tag of nine characters — with the base relation ATR depends on a noun or a pronoun (n or p in the first position) and does not agree with it in gender, number or case. The head is the effective head (defined at the beginning of S2), so that a conjunct is compared with what the coordination depends on. Two values are compatible when they are equal or when either of them is the hyphen; the common gender c is compatible with every gender, and the dual with the plural. Participles, which in the AGDT tagset are verbs with the mood p, are not examined. The error may lie in the modifier or in the head.

Parameters. None.

Exclusions. Modifiers in the genitive, which may be adnominal genitives rather than agreeing attributes; elliptic modifiers and heads, as defined under double-accent; the quantifiers of the list declared in S6; the indeclinable numerals of the list declared in S6, both as modifiers and as heads.

Outcome. Case to check, for manual review.

case-object — Object in an unexpected case
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: form, lemma, morphological tag, relation, head and identifier.

Condition. A noun, an adjective, a pronoun, an article or a numeral — n, a, p, l or m in the first position of a tag of nine characters — has the base relation OBJ and is marked as nominative or vocative. Since OBJ denotes any argument of the governor (Celano 2014, § 3.3), case, relation and head must be checked together. Two classes of tokens are not reported: the verbal adjectives in -τέος, -τέα, -τέον, recognised when the form or the lemma, longer than four characters, ends in -τέον, -τέος, -τέα, -τέου, -τέῳ or -τέων (Table S4.2); and the heads of a clause, recognised because at least one token with the base relation SBJ, OBJ or PNOM has them as its effective head.

Parameters. None.

Exclusions. The verbal adjectives and the clause heads just described.

Outcome. Case to check, for manual review.

case-subject — Subject in a case unexpected for the context
Data examined. Imported tokens, sentence by sentence, technical tokens included. Fields read: morphological tag, relation, head and identifier.

Condition. A noun, an adjective, a pronoun, an article or a numeral — n, a, p, l or m in the first position of a tag of nine characters — has the base relation SBJ and a case other than the nominative; the hyphen and the underscore in the position of case are not cases. The AGDT 2.0 guidelines admit a subject in the genitive in the genitive absolute and in the accusative in the infinitive clause or with the supplementary participle (Celano 2014, § 3.2). The program therefore does not report the subject when the tags show one of these constructions: an accusative subject whose head is an infinitive, or a subject whose head is a participle in the same case, which also covers a dative subject under a dative participle. The head is looked for in three places: the written head; the effective head (defined at the beginning of S2); and the conjuncts of the written head, that is, its dependents whose relation ends in _CO, as when the subject depends on the coordinator of two infinitives or of two participles. The subjects for which neither the written head nor the

effective head has a tag of nine characters beginning with a value of the AGDT inventory are not reported either: this happens when the subject is attached to the root, or to a node without a tag, such as an elliptic node. What remains is reported with one of two explanations: a genitive or an accusative outside the constructions that license it, where the case or the head must be checked, or a case that the guidelines never admit for the subject, such as the dative or the vocative.

Parameters. None.

Exclusions. None.

Outcome. Case to check, for manual review.

rare-relation-for-lemma-and-pos — Rare relation for lemma and POS
Data examined. Words, grouped by lemma and part of speech. Fields read: lemma, part of speech and relation.

Condition. The words are grouped by lemma and part of speech. For every group of at least two hundred words in the run, the program computes the distribution of the base relations (defined at the beginning of S2) and reports the words whose base relation accounts for at most one per cent of the group. The lemma is compared in NFC without removing the final number; the part of speech is the first position of a nine-character tag, when it is a value of the AGDT inventory. Rarity does not prove an error: a construction may be rare and correct, and the absolute number shows how much weight the percentage carries.

Parameters. Minimum support 200 occurrences; maximum share 1%.

Exclusions. Words with an empty lemma, part of speech or relation; the Pedalion value b counts as an empty part of speech, so that the words that carry it are not examined.

Outcome. Case to check, for manual review.

S3. The authorities of the criterion

Every check rests on a declared authority: what makes its criterion verifiable independently of the authors (§ 4.2 of the article). The five authorities, with the number of checks that rest on each, are those of Table 4 of the article, repeated in Table S3.1; Table S3.2 gives the authority of each of the 77 checks, with its source. The program does not need this information to run, and the file data/catalogo_controlli.csv does not contain it.

The attribution follows five rules, applied in the same way to all 77 checks. (1) Structure of the data: the defect concerns the well-formedness of the token or of the tree and is recognisable without reading the Greek. (2) AGDT guidelines: the condition compares a value or a relation with the inventory of the scheme or with one of its explicit rules; the references are Bamman and Crane (2008) for syntax and Celano (2014) for morphology. (3) Lexica and grammars: the condition rests on a linguistic fact documented in LSJ or in Smyth, directly or through a closed list (S6). (4) Analysed corpus: what makes the finding verifiable is the way in which the resource itself writes, so that the finding can be confirmed or refuted only by observing the resource. (5) Project convention: the resource declares the usage in its own documentation.

A numerical threshold does not by itself make the corpus the authority. Six of the eleven checks founded on the analysed corpus use none of the parameters of S4 — apostrophe-mixed, unicode-not-nfc, lemma-tool-numbering, lemma-graphic-variant, nonlexical-z-pos and case-letter-l — and conversely relation-truncated-or-foreign uses two thresholds but rests on the AGDT inventory (see below). The locative of case-letter-l is admitted by the AGDT inventory (Celano 2014, § 2): what makes it worth reporting is its rarity in the resource, and this is why its authority is the analysed corpus.

Project convention is the authority of one check only, pedalion-b-pos: the README of the Pedalion Trees declares the part of speech b. Conventions that a resource applies without declaring them, such as the Perseus lemma numbering or the systematic bare label CO, rest on the analysed corpus. GLAUx declares its NFD encoding ("the GLAUx corpus is encoded in Unicode-NFD, meaning that diacritics are separate characters": GLAUx README); corpus-not-nfc nevertheless rests on the analysed corpus, because its criterion — the share of tokens not in NFC — is verified by observing the resource and applies equally to resources that declare nothing.

Table S3.1 The five authorities, with the number of checks that rest on each.

Authority	Checks	What must be true	How it is refuted
AGDT guidelines	29	the scheme declares the rule applied	by citing the point of the guidelines that says otherwise
structure of the data	21	the defect is recognisable without reading the Greek	by showing that the structure admits that case
lexica and grammars	15	the linguistic property is documented	by citing the lexicon or the grammar
analysed corpus	11	the internal distribution is stable	by showing that the distribution does not hold
project convention	1	the project declares the convention	by showing that the declaration is not there
Total	77		

Table S3.2 The principal authority of each of the 77 checks, with the source of, or the reason for, the attribution. The order is that of Table S1.1.

Identifier	Principal authority	Source or reason for the attribution
creon-anomaly	lexica and grammars	LSJ s.v. χρεών: neuter noun, used mostly in the nominative and accusative
dein-participle	lexica and grammars	LSJ s.v. δεῖ (infinitive δεῖν, participle δέον); Smyth § 397 a
deina-anomaly	lexica and grammars	LSJ s.v. δεῖνα; Smyth § 336: the indefinite pronoun δεῖνα
invalid-character	AGDT guidelines	inventory of the values admitted for each position
invalid-length-broken	AGDT guidelines	the tag has nine positions
invalid-length-padding	AGDT guidelines	the tag has nine positions
leading-t	AGDT guidelines	inventory of the parts of speech
leading-x-lexical	AGDT guidelines	x reserved for elliptic nodes
missing-postag	structure of the data	obligatory field absent or empty
numeral-generic-m	AGDT guidelines	positional encoding: the underscore is not a value of the inventory; the tag of a numeral without number, gender and case is m-------- (Smyth § 350: indeclinable numerals)
osfranthiti-anomaly	lexica and grammars	Smyth, list of verbs, s.v. ὀσφραίνομαι (aorist passive ὠσφράνθην); § 125 b: imperative in -θητι
participle-mood-g	AGDT guidelines	inventory of mood: i, s, o, n, m, p
pot-anomaly	lexica and grammars	Smyth §§ 181 b, 346: interrogative πότε; §§ 70, 124: elision and aspiration before the rough breathing
punctuation-leading-e	AGDT guidelines	encoding of punctuation: u in the first position
san-anomaly	lexica and grammars	LSJ s.v. Σ σ ς: σάν, τό, Doric name of sigma (Hdt. 1.139)
slot-forbidden	AGDT guidelines	features admitted for each part of speech
slot-missing	AGDT guidelines	features required for each part of speech
tessereskaideka-anomaly	lexica and grammars	Smyth §§ 347 D.2, 349 g: Ionic τεσσερεσκαίδεκα, indeclinable
missing-lemma	structure of the data	obligatory field absent or empty
apos-suffix-orphan	AGDT guidelines	rule of the suffix _AP
coord-suffix-orphan	AGDT guidelines	rule of the suffix _CO
head-cycle	structure of the data	the chain of heads returns to a node already traversed
head-dangling	structure of the data	the declared head does not exist in the sentence

Identifier	Principal authority	Source or reason for the attribution
head-missing	structure of the data	head field empty where the sentence is annotated
relation-greek-lookalike	AGDT guidelines	inventory of the syntactic relations
relation-missing	structure of the data	obligatory field absent or empty
relation-nil-placeholder	structure of the data	placeholder nil in place of the value
relation-truncated-or-foreign	AGDT guidelines	inventory of the syntactic relations: the label does not appear in it, and the two diffusion thresholds decide only which of the two twin checks fires
relation-typo	AGDT guidelines	inventory of the syntactic relations
relation-unknown	AGDT guidelines	inventory of the syntactic relations
sentence-no-root	AGDT guidelines	an AGDT sentence has a root
sentence-not-annotated	structure of the data	no syntactic link declared in the sentence
token-id-duplicate	structure of the data	two tokens of the same sentence with the same identifier
annotator-note-in-relation	structure of the data	working text inside a data field
beta-code-form	structure of the data	transliteration residue inside a Unicode field
beta-code-lemma	structure of the data	transliteration residue inside a Unicode field
breathing-impossible	lexica and grammars	Smyth §§ 9, 13: breathings only on initial vowels, diphthongs and ρ; § 70: the apostrophe marks the elided vowel
final-sigma-inside	structure of the data	final sigma in internal position
invisible-char-form	structure of the data	Unicode format character inside the value
invisible-char-lemma	structure of the data	Unicode format character inside the value
isolated-breathing	structure of the data	the form consists of the diacritic alone
mixed-script	structure of the data	two alphabets within the same string
mixed-script-unrepairable	structure of the data	structure of the data as defined by GCP: a Greek form is written in a single script (see the note after the table)
whitespace-in-form	structure of the data	separator inside a value that must be a single token
whitespace-in-lemma	structure of the data	structure of the data as defined by GCP: a lemma is a single citation form, without internal separators (see the note after the table)
degree-underscore	AGDT guidelines	positional encoding: the hyphen for the unfilled feature
grade-p	AGDT guidelines	inventory of degree: c and s
pedalion-b-pos	project convention	README of the Pedalion Trees, which declares the part of speech b
personal-pronoun-person	AGDT guidelines	AGDT 2.0 does not require person on the pronoun (Celano 2014, § 2.3); the criterion of the report is the profile adopted by GCP (see the note after the table)
underscore-person-number	AGDT guidelines	positional encoding: the hyphen for the unfilled feature
lemma-graphic-variant	analysed corpus	two spellings used by the same resource for the same headword (list varianti-canoniche-di-lemma, S6)
lemma-suspicious-characters	structure of the data	non-alphabetic characters inside the lemma
lemma-tool-numbering	analysed corpus	numeral that the resource appends to the lemma
apostrophe-mixed	analysed corpus	coexistence of several code points for the same sign within the same file
corpus-not-nfc	analysed corpus	share of tokens not in NFC equal to or greater than 50% of the active tokens of the resource
unicode-not-nfc	analysed corpus	tokens not in NFC within a resource that is predominantly in NFC
case-letter-l	analysed corpus	the locative is in the AGDT inventory (Celano 2014, § 2): the check reports every occurrence, and the reason for reporting a value that the scheme admits is its rarity in the resource
common-gender-c	AGDT guidelines	inventory of genders: m, f, n
dein-participle-disputed	lexica and grammars	LSJ s.v. δεῖ, IV: participial δεῖν doubtful in Ar. Fr. 220 and Lys. 14.7
mediopassive-d	AGDT guidelines	inventory of voice: a, p, m, e
nonlexical-z-pos	analysed corpus	use of z as a marker specific to the resource (not executable with the adopted filter, S2)
slot-missing-bare	AGDT guidelines	features required for the noun
tan-anomaly	lexica and grammars	LSJ s.v. τᾱ́ν (ὦ τᾱ́ν); Smyth § 332 D: the Doric forms of the article

Identifier	Principal authority	Source or reason for the attribution
verbal-adjective-g	lexica and grammars	Smyth §§ 471–473, 2149–2152: the verbal adjectives in -τέος
lemma-rare-for-form	analysed corpus	thresholds 25 · 1 · 95% on the distribution of lemmas by form and part of speech
uninflected-lemma	analysed corpus	lemma with at least five nominal occurrences, none of which carries case
aux-finite-verb	AGDT guidelines	use of AuxY and AuxZ
aux-governor-without-child	AGDT guidelines	use of AuxP and AuxC
aux-governs-conjuncts	AGDT guidelines	use of _CO and of COORD
coord-discordant-conjuncts	AGDT guidelines	use of _CO in coordination
relation-bare-convention	analysed corpus	at least 20 occurrences and presence in at least 90% of the documents of the resource
sentence-many-roots	AGDT guidelines	an AGDT sentence has a single root
double-accent	lexica and grammars	Smyth §§ 181–185: the enclitics and the accent they throw back
agreement-modifier	lexica and grammars	Smyth § 1020: concord of the adjective, the article and the participle (the check examines articles, adjectives and numerals: S2)
case-object	lexica and grammars	Smyth § 1281: the cases of the predicate (cf. §§ 1339, 1341)
case-subject	lexica and grammars	Smyth §§ 927, 938–939: the subject in the nominative (exceptions §§ 1972, 2070, 2076)
rare-relation-for-lemma-and-pos	analysed corpus	thresholds 200 · 1% on the distribution of relations by lemma and PoS

As Table S3.2 shows, none of the eleven checks founded on the analysed corpus is a verifiable violation: the 45 violations rest on the structure of the data (20), on the AGDT guidelines (17) and on the lexica and grammars (8). A phenomenon that is rare, or regular, within a corpus is never for that reason an error (§ 4.2 of the article). The bare relation label shows how threshold and authority are kept apart. Below the two diffusion thresholds of S4 the label is reported by relation-truncated-or-foreign, a verifiable violation that rests on the AGDT inventory, which does not contain the label; above them it is reported by relation-bare-convention, a case to check that rests on the analysed corpus. The thresholds decide which of the two checks fires, not what makes the criterion verifiable. In the same way the scope of slot-missing, but not its criterion, depends on a count made over the run: the occurrences of the lemmas that the corpus never inflects pass to uninflected-lemma.

Five attributions call for a word of explanation. personal-pronoun-person: the AGDT 2.0 guidelines annotate the pronoun for gender, number and case (Celano 2014, § 2.3), so that a pronoun without person never violates the scheme; the check registers it as a systematic variant with respect to the profile adopted by GCP, which fills person on personal pronouns (§ 6.3 of the article). numeral-generic-m: the condition is the single tag m-_---__-, whose underscores in number, gender and case are not values of the AGDT inventory; the 28 tokens concerned are numerals written with letters or indeclinable cardinals, which carry no number, gender or case (Smyth § 350), and the program proposes m--------. The underscore as such is registered, wherever it occurs, by underscore-person-number, which on these 28 tokens proposes the same tag. The two checks thus give the same 28 tokens different classes, and both findings are counted: underscore-person-number registers how the resource writes a position, whatever the word, while numeral-generic-m, like the checks on the tags of χρεών, δεῖνα and σάν (creon-anomaly, deina-anomaly, san-anomaly), reports that the tag of a known word leaves undetermined what the grammar determines: on these numerals, that number, gender and case do not apply. If these 28 findings were counted as variants, the verifiable violations would fall from 1,914 to 1,890 in Gorman (31.0 to 30.6 per 10,000 active tokens) and from 1,361 to

1,357 in Pedalion (49.6 to 49.4), and their share of the findings would remain 1.9% and 4.6%. whitespace-in-lemma and mixed-script-unrepairable: the rule is a rule of the data adopted by GCP — a lemma is a single citation form, a Greek form is written in one script — and the finding says that the token breaks it, not that the reading is wrong. Of the four lemmas with an internal space, two are compounds split into their parts (καί τοι for καίτοι in Gorman, πρὸ - ἔργον for προὔργου in Pedalion), one is a token of several words (εἴκοσι καὶ τετρακοσίων in Pedalion) and one is a damaged lemma of Μακεδόνιος, in which a space carrying the acute accent stands in place of the accented omicron (first GLAUx sample). Of the fourteen forms in two scripts, the eleven of Pedalion are in the file of papyri, where a Latin G marks a lost letter (κGματα), and the three of GLAUx attach the lacuna marker G? to the adjacent words (τῶνG?κατὰ): in all of them the correction requires an editorial decision, and for this reason the checks propose none. breathing-impossible: breathings are written only on initial vowels and diphthongs and on ρ (Smyth §§ 9, 13), and the place of an elided vowel is marked by the apostrophe (Smyth § 70); a smooth breathing on a final consonant, as in δ̓ for δ’, stands for the apostrophe. The 19 findings, all in Gorman, are the form δ̓ with the lemma δέ.

S4. The nine numerical parameters

The nine numerical values chosen by the authors (§ 4.3 of the article) are listed in Table S4.1. Unlike the authorities of S3, they do not come from a source: they turn a general criterion into an applicable rule. For example, a relation is rare for a pair of lemma and part of speech when the pair occurs at least 200 times and the relation accounts for at most 1% of its occurrences. The values are written in the code. Thresholds fixed by the scheme, such as the nine characters of the morphological tag, are not parameters in this sense.

Table S4.1 The nine parameters, with the value adopted, the module in which it is written and the check that uses it.

Parameter	Value	Module	Check	What it decides
RARE_RELATION_MIN_SUPPORT	200	agreement.py	rare-relation-for-lemma-and-pos	how many occurrences the pair of lemma and part of speech must have for the check to examine it
RARE_RELATION_MAX_SHARE	0.01	agreement.py	rare-relation-for-lemma-and-pos	the maximum share of the group’s occurrences, inclusive, at which a relation is called rare
min_form_total	25	lemma.py	lemma-rare-for-form	how many occurrences the pair of form and part of speech must have for the check to examine it
max_rare	1	lemma.py	lemma-rare-for-form	how many occurrences the minority lemma may have: only one, and this is the definition of the check, not a statistical threshold

Parameter	Value	Module	Check	What it decides
min_major_share	0.95	lemma.py	lemma-rare-for-form	how much the majority lemma must cover for the isolated occurrence to stand out
WHOLE_CORPUS_ENCODING	0.50	orthography.py	corpus-not-nfc	the share of active tokens not in NFC, inclusive, from which the encoding belongs to the resource and is reported only once
INDECLINABLE_MINIMUM	5	postag.py	uninflected-lemma	how many nominal occurrences are needed for a lemma that never carries case to count as a lemma that the corpus does not inflect: from that number upwards its occurrences without case are reported once per lemma by uninflected-lemma, and removed from slot-missing and slot-missing-bare
BARE_RELATION_MINIMUM	20	syntax.py	relation-bare-convention and relation-truncated-or-foreign	how many occurrences are needed for the bare label to qualify as a possible convention rather than an oversight
BARE_RELATION_SPREAD	0.90	syntax.py	relation-bare-convention and relation-truncated-or-foreign	in what proportion of the documents it must occur to be a convention of the resource

The nine values serve six checks: lemma-rare-for-form uses three of them; rare-relation-for-lemma-and-pos, relation-bare-convention and relation-truncated-or-foreign two each; corpus-not-nfc and uninflected-lemma one each. The two thresholds of the bare relation label belong to both twin checks, because they decide, token by token, which of the two fires. The thresholds of corpus-not-nfc and of uninflected-lemma belong only to the check that reports the phenomenon as a whole once the threshold is reached; unicode-not-nfc, slot-missing and slot-missing-bare, which report token by token what remains below it, receive it as an exclusion.

The two thresholds of the bare relation label concern the second phenomenon of § 6.3 of the article. A label written without the function that should precede it, such as CO, is treated as a systematic fact of the resource if it occurs at least 20 times and in at least 90% of the documents; then it is reported by relation-bare-convention as a case to check, without a proposed correction. Below either threshold it is reported by relation-truncated-or-foreign as a verifiable violation. The

thresholds are evaluated label by label on the imported tokens of the run, technical tokens included. In the data they separate the automatic strata of the two GLAUx samples, where bare CO occurs 2,369 and 1,887 times among the imported tokens, in all eighteen and all seventeen documents, from Gorman and Pedalion, where it occurs 14 times in 6 documents out of 158 and 32 times in 10 out of 40. S7.7 relates these counts, made on the imported tokens, to the counts of findings given in § 6.3 of the article.

Besides the nine parameters, a few fixed limits in the code decide which strings or structures a check considers at all; they are not exposed as parameters and are not varied. Table S4.2 lists those that affect the checks. Three of them change the findings of the six runs: without the limit of five capital letters, slot-missing would report 351 tokens instead of 356 in the automatic stratum of the first GLAUx sample; without the minimum length of the core, final-sigma-inside would also report the Pedalion form ϛZ; without the minimum excess of length of the host word, double-accent would report one token fewer in Gorman and five fewer in Pedalion.

Table S4.2 Fixed numerical limits in the code that delimit the scope of some checks. They are not parameters in the sense of S4: they are not varied, and they are not attributed to an authority.

Limit	Module	Checks	Effect in the six strata
a form or lemma, compared in NFC, written in Greek letters or numeral signs without accents, breathings, iota subscript or diaeresis counts as a numeral written with letters, or as a diagram label, if it ends with a keraia (or a sign used in its place), if it consists of a single character or if its letters, at most five, are all capitals	postag.py	slot-missing, slot-missing-bare, uninflected-lemma (exclusion)	changes the findings: slot-missing reports 356 tokens in the automatic stratum of the first GLAUx sample, and would report 351 without the limit
final sigma: values shorter than two characters, and cores of at most two characters, are not examined	orthography.py	final-sigma-inside	changes the findings: excludes the Pedalion form ϛZ
the search for the effective head of a node stops after twelve steps	syntax.py	coord-suffix-orphan, apos-suffix-orphan, coord-discordant-conjuncts, agreement-modifier, case-subject, case-object	none: the limit is never reached
an enclitic written together with its host is recognised when the host is longer than the enclitic by more than two characters	orthography.py	double-accent (exclusion)	changes the findings: without the limit, double-accent would report 54 instead of 55 tokens in Gorman and 19 instead of 24 in Pedalion
an annotator's note in the relation field: two or more of the signs ! ? %, or four or more lower-case Latin letters followed by a space	orthography.py	annotator-note-in-relation; relation-unknown (exclusion)	not measured
verbal adjectives excluded from case-object: forms or lemmas longer than four characters	agreement.py	case-object (exclusion)	none in the six strata
placeholder forms recognised by the diagnostic checks: [0] to [20] (the filter of S11.1 accepts any number)	postag.py, agdt_schema.py, orthography.py	missing-postag, leading-x-lexical, invalid-character, double-accent, agreement-modifier	none: the largest placeholder in the data is [16]
tags that the padding can repair: eight characters, or more than nine	postag.py	invalid-length-padding, invalid-length-broken	part of the definition of the two checks (S2)

S5. The sensitivity analysis for the thresholds

Four parameters of two checks were varied (§ 4.3 of the article): the minimum support and the maximum share of rare-relation-for-lemma-and-pos, and the minimum support and the minimum majority of lemma-rare-for-form. Fifteen configurations, the adopted ones included, were evaluated on the six strata, for 90 evaluations (Tables S5.1 and S5.2). When the parameters move in the same direction, the more permissive configuration keeps all the findings of the adopted one and may add others, and the more restrictive one keeps part of them. Only one evaluation both loses and adds findings: support 50 with majority 90% on Gorman (42 lost, 5 new), where the two parameters move in opposite directions.

The other parameters were not varied. The maximum number of rare occurrences of lemma-rare-for-form is fixed at one because it is the definition of the check, not a statistical threshold (S5.2). The four remaining parameters — the 50% share of corpus-not-nfc, the five occurrences of uninflected-lemma and the two diffusion thresholds of the bare label — do not choose how many findings to produce along a graded distribution: they separate two qualitative situations and decide which of two twin checks fires, and this section says nothing about their sensitivity. The adopted values are those of the release with which the six runs were carried out.

The evaluations were computed from the tokens of the six runs (S11), by calling the two checks with the alternative arguments; the two adopted configurations reproduce exactly the findings of the two checks in the six runs. A finding is identified by source file, document, sentence index and token index. Each cell of Tables S5.1 and S5.2 gives, in order, the findings of the configuration, those of the adopted configuration that it loses and those that it adds; a configuration is nested in, or contains, the adopted one when one of the last two numbers is zero.

S5.1 rare-relation-for-lemma-and-pos

Six configurations — support 100 or 200 crossed with share 0.5%, 1% or 2% — on the six strata: 36 evaluations. Nesting holds along both axes: at fixed support, share 0.5% is contained in 1% and 1% in 2%; at fixed share, support 200 is contained in support 100. The more permissive configurations keep 100% of the adopted findings; the more restrictive ones keep between 24.5% and 73% of them.

Table S5.1 rare-relation-for-lemma-and-pos: the 36 configuration × stratum evaluations. In each cell: findings · lost · new with respect to the adopted configuration (support 200, share 1%).

Configuration	Gorman	Pedalion	GLAUx auto.	GLAUx man.	Conf. auto.	Conf. man.
support 100, share 0.5%	1,548 · 1,217 · 0	641 · 277 · 0	308 · 209 · 0	6 · 18 · 0	202 · 622 · 0	27 · 10 · 0
support 100, share 1%	2,950 · 0 · 185	1,034 · 0 · 116	632 · 0 · 115	31 · 0 · 7	930 · 0 · 106	43 · 0 · 6
support 100, share 2%	5,805 · 0 · 3,040	2,059 · 0 · 1,141	2,226 · 0 · 1,709	62 · 0 · 38	2,084 · 0 · 1,260	121 · 0 · 84
support 200, share 0.5%	1,548 · 1,217 · 0	641 · 277 · 0	308 · 209 · 0	6 · 18 · 0	202 · 622 · 0	27 · 10 · 0
support 200, share 1% (adopted)	2,765 · 0 · 0	918 · 0 · 0	517 · 0 · 0	24 · 0 · 0	824 · 0 · 0	37 · 0 · 0
support 200, share 2%	5,327 · 0 · 2,562	1,834 · 0 · 916	1,941 · 0 · 1,424	45 · 0 · 21	1,829 · 0 · 1,005	105 · 0 · 68

The share decides and the support adjusts. On Gorman, at share 1%, lowering the support from 200 to 100 adds 6.7% to the findings; at support 200, raising the share from 1% to 2% adds 92.7% and lowering it to 0.5% removes 44.0%. Gorman and Pedalion, the two large manually annotated corpora, are the most stable: at support 200, over a range of the share from half to double the

adopted value, they move between −44% and +93% and between −30% and +100%, whereas the automatic stratum of the first GLAUx sample reaches +275%. Over the whole of Table S5.1 the largest change for Gorman and Pedalion is that of support 100 with share 2%: +109.9% and +124.3%. The two manual strata of GLAUx are small — 24 and 37 findings — and their variation reflects the size of the sample, not the check.

The counts grow steadily with the share. At support 200 Gorman gives 1,047, 1,548, 1,810, 2,765, 3,263, 3,462, 5,327, 6,194 and 8,808 findings at shares of 0.25%, 0.5%, 0.75%, 1%, 1.25%, 1.5%, 2%, 2.5% and 3%. The adopted value, 1%, lies in the middle of the range in which the check still indicates rarity: at 0.25% Gorman loses 62% of its findings and the manual stratum of the first GLAUx sample keeps none; at 3% Gorman triples them.

S5.2 lemma-rare-for-form

Nine configurations — support 15, 25 or 50 crossed with majority 90%, 95% or 98% — on the six strata: 54 evaluations. The maximum number of rare occurrences is fixed at one, because the check is defined as a lemma that appears “on a single occasion” for that form; varying it would ask a different question. Nesting holds along each axis. Only one evaluation both loses and adds findings — support 50 with majority 90% on Gorman, 42 lost and 5 new — and in it the two parameters pull in opposite directions.

Table S5.2 lemma-rare-for-form: the 54 configuration × stratum evaluations. In each cell: findings · lost · new with respect to the adopted configuration (support 25, majority 95%). The only evaluation with both lost and new findings is Gorman with support 50 and majority 90%.

Configuration	Gorman	Pedalion	GLAUx auto.	GLAUx man.	Conf. auto.	Conf. man.
support 15, majority 90%	186 · 0 · 84	49 · 0 · 8	0 · 0 · 0	2 · 0 · 0	0 · 0 · 0	0 · 0 · 0
support 15, majority 95%	124 · 0 · 22	44 · 0 · 3	0 · 0 · 0	2 · 0 · 0	0 · 0 · 0	0 · 0 · 0
support 15, majority 98%	54 · 48 · 0	27 · 14 · 0	0 · 0 · 0	1 · 1 · 0	0 · 0 · 0	0 · 0 · 0
support 25, majority 90%	116 · 0 · 14	41 · 0 · 0	0 · 0 · 0	2 · 0 · 0	0 · 0 · 0	0 · 0 · 0
support 25, majority 95% (adopted)	102 · 0 · 0	41 · 0 · 0	0 · 0 · 0	2 · 0 · 0	0 · 0 · 0	0 · 0 · 0
support 25, majority 98%	54 · 48 · 0	27 · 14 · 0	0 · 0 · 0	1 · 1 · 0	0 · 0 · 0	0 · 0 · 0
support 50, majority 90%	65 · 42 · 5	28 · 13 · 0	0 · 0 · 0	1 · 1 · 0	0 · 0 · 0	0 · 0 · 0
support 50, majority 95%	60 · 42 · 0	28 · 13 · 0	0 · 0 · 0	1 · 1 · 0	0 · 0 · 0	0 · 0 · 0
support 50, majority 98%	54 · 48 · 0	27 · 14 · 0	0 · 0 · 0	1 · 1 · 0	0 · 0 · 0	0 · 0 · 0

Here too the counts change steadily. For the majority, the largest step lies between 96% and 97%, where Gorman loses 31 findings, and the adopted 95% lies on the flat stretch before it: 116, 112, 107, 102 and 101 findings at 90%, 92%, 94%, 95% and 96%. For the support, the decline from 25 onwards is regular: 102, 86, 68, 60, 55 and 46 findings at 25, 30, 40, 50, 75 and 100.

Since at most one rare occurrence is allowed, every finding is a single token: the 102 findings of Gorman are 102 distinct cases. Three strata give zero in all nine configurations — the automatic strata of the two samples and the manual stratum of the confirmation sample — although they have from 61 to 1,587 eligible pairs, depending on the stratum and on the support: the zero is not an effect of the thresholds. In these strata almost every frequent pair of form and part of speech has a single lemma: at support 25, only 3 of 892, 2 of 842 and 2 of 107 pairs have more than one, and none of these has an isolated lemma. This does not mean that automatic lemmatisation is consistent by construction: the two automatic strata have 47 and 10 variation nuclei at the level of the lemma (Table S8.1).

In neither analysis do the verifiable violations move: the two checks varied are cases to check, and no violation depends on the four parameters varied. Three of the parameters not varied, by contrast, also affect checks of the violation class — the five occurrences of uninflected-lemma, which remove from slot-missing and slot-missing-bare the lemmas that the corpus never inflects, and the two diffusion thresholds, which decide when the bare label is a violation — and their sensitivity has not been measured.

S6. The ten closed lists

The program distributes ten closed lists (§ 4.3 of the article). Six are read by diagnostic checks: enclitiche, loci-dein-participiale, numerali-indeclinabili, pronomi-personali, quantificatori-non-concordanti and varianti-canoniche-di-lemma. The other four enter no finding: lemmi-grammaticali and lemmi-particella serve the statistics, risorse-con-metadati the recognition of the resources for which the program distributes metadata, and estensioni-locali-di-pos documents the value b declared by Pedalion, which the program writes directly in the code of pedalion-b-pos. Table S6.1 gives, for each list, its size, the checks that read it and its source, which is also declared in the file of the list. The lists are in NFC, and the program compares them with the tokens in NFC, so that they apply to a resource in NFD as well.

Table S6.1 The ten closed lists distributed with the program, the number of entries, the consumer and the source.

List	Entries	Read by	Source
enclitiche	82	double-accent	grammar: Smyth §§ 181–185 (with § 181 D), § 334 D, § 346 D.1; four entries are not enclitics in Smyth: ἄττα (explicitly not enclitic, § 334 a), μεν, μεο, σφεσι
estensioni-locali-di-pos	1	no part of the program: the value b is written in the code	Pedalion README for b; the value z does not appear in it, because it marks a technical token and not a category of words
loci-dein-participiale	2	dein-participle and dein-participle-disputed	LSJ s.v. δεῖ, IV: participial δεῖν doubtful in Ar. Fr. 220 and Lys. 14.7
numerali-indeclinabili	47	slot-missing, slot-missing-bare, uninflected-lemma, agreement-modifier	Smyth § 350 (the cardinals from 5 to 199 are indeclinable; cf. §§ 347 D, 349 d, g, D); LSJ s.vv. πέντε, ἑκατόν, δύο; δύο is only optionally indeclinable (§ 349 d), and in τρεῖς καὶ δέκα the first member is inflected (§ 349 g)
pronomi-personali	3 groups (36 forms)	personal-pronoun-person	grammar: Smyth § 325 (with § 325 a, d, D); the list does not contain the unaccented μου, μοι, σου, σοι, οὑ, οἱ nor the dual σφώ, σφῷν
quantificatori-non-concordanti	4	agreement-modifier (exclusion)	grammar: Smyth §§ 952, 962 a, 982, 999 for ἄμφω, ἑκάτερος, ἕκαστος; no statement in Smyth has been found for εἷς

List	Entries	Read by	Source
lemmi-grammaticali	136	no check: the “Lexical words only” view and the frequencies	compiled by hand from a survey of the three corpora
lemmi-particella	56	no check: the count of particles in the statistics	compiled by hand
risorse-con-metadati	6	no check: the recognition of the resource by fingerprint	fingerprints computed on the imported corpus, before any filter
varianti-canoniche-di-lemma	12	lemma-graphic-variant; outside the diagnosis, the comparison between different corpora	reading of the corpora

One list and two parameters concern the two phenomena of § 6.3 of the article. Person on the personal pronoun depends on the list pronomi-personali, which decides which tokens personal-pronoun-person examines (case study in S7.6). The bare relation label CO depends not on a list but on the two diffusion thresholds of S4, which decide between relation-truncated-or-foreign and relation-bare-convention (case study in S7.7). The two phenomena rest on different authorities (S3): the pronoun on the AGDT guidelines, which do not require person; the bare label on the AGDT inventory when it is sporadic and on the analysed corpus when it is systematic.

Of the six lists read by the checks, five rest on the grammar (enclitics, personal pronouns, quantifiers), on grammar and lexicon (numerals) or on the lexicon (the passages of δεῖν), the sources of the authority “lexica and grammars” of S3; varianti-canoniche-di-lemma rests on the reading of the corpora. The four that rest wholly or partly on the grammar do not coincide entry by entry with Smyth (1920), as Table S6.1 shows: four of the 82 enclitics are not enclitics in Smyth, the personal pronouns lack some unaccented and dual forms, Smyth says nothing of εἷς as a quantifier exempt from agreement, and two of the 47 numerals are indeclinable only in part of their uses. The lists are distributed as the program uses them. The effect of each difference was measured by running the checks that read the lists again on the six strata, with one list aligned with Smyth at a time (Table S6.2). No proposed correction changes, because none of the checks concerned proposes one; all the changes together would add nine verifiable violations (slot-missing: eight in Gorman, one in Pedalion), so that the rate per 10,000 active tokens would go from 31.0 to 31.2 in Gorman and remain the same in the other five strata.

Table S6.2 Findings added when one list is aligned with Smyth. The two manual strata of GLAUx, not shown, give no change; no proposed correction changes.

List and change	Check affected (class)	Gorman	Pedalion	GLAUx first sample, automatic	GLAUx confirmation, automatic
enclitiche: remove ἄττα, μεν, μεο and σφεσι	double-accent (case to check)	+1	0	0	+1
pronomi-personali: add μου, μοι, σου, σοι, οὐ, οἱ, σφώ and σφῷν	personal-pronoun-person (systematic variant)	+397	+112	+1,169	+844
quantificatori-non-concordanti: remove εἷς	agreement-modifier (case to check)	+13	+9	+15	+7
numerali-indeclinabili: remove δύο	slot-missing (verifiable violation)	+8	+1	0	0

List and change	Check affected (class)	Gorman	Pedalion	GLAUx first sample, automatic	GLAUx confirmation, automatic
	slot-missing-bare (case to check)	+77	+6	0	0
	agreement-modifier (case to check)	+166	+4	0	0
numerali-indeclinabili: remove τρεῖς καὶ δέκα	—	0	0	0	0

S7. Ten case studies

Ten case studies follow, chosen among the checks with the most findings and among those that best show the difference between the three diagnostic classes. Each gives a real occurrence, with its document and sentence, the counts on the six strata, the class, the treatment and the reason for the classification. S7.6 and S7.7 are the two phenomena of § 6.3 of the article; S7.9 and S7.10 take up, with all the counts, two examples of § 5.2.

S7.1 The numeral appended to the lemma (Gorman)

Gorman, Aeschines, *Against Timarchus* (aeschines-1-1-50-bu1), sentence 1, token 5: the comma carries the lemma punc1. The final digit is not a Greek letter: it is the number with which the Perseus project distinguishes homographic entries of the lexicon (λέγω1, λέγω2, λέγω3), and the same convention lemmatises punctuation as punc1. As long as it remains, the same word is counted twice in any lexical comparison between resources. The check fires 72,836 times in Gorman — 62,238 on punctuation and 10,598 on words — and 311 times in Pedalion; it fires in none of the four strata of GLAUx. It is a systematic variant with manual review: the program does not remove the digit by itself, because removing the number from λέγω3 means deciding that that entry and λέγω1 are the same, and the decision is a lexicographic one. On its own, this convention produces 79.2% of the systematic variants of Gorman and 72.1% of all its findings. The AGDT 2.0 guidelines prescribe that such numbers be removed from the lemma field (Celano 2014, § 2.5), and in the AGDT 2.1 release the lemma of a punctuation mark is the mark itself (AGDT 2.1 README); Gorman follows the AGDT 1.1 tagset, whose guidelines contain no rule on lemmas, and keeps the numbers regularly: a value used with regularity in the corpus is recorded as a variant (§ 3.2 of the article). The same problem is noted by Vatri and McGillivray (2020: 185) in their evaluation of Greek lemmatisers, where such numbers were disregarded “since lexica used by different lemmatizers are not expected to be aligned”.

S7.2 The underscore in the degree position (Gorman)

Gorman, same document, sentence 1, token 36: μεγάλα, lemma μέγας, tag a-p---na_. The ninth position contains an underscore instead of the hyphen. The two signs say the same thing — here the feature is not filled — but for a program they are different symbols, and the same analysis ends up counted as two different tags. The check fires 11,187 times in Gorman and 270 in Pedalion. It is a systematic variant with automatic correction, because the replacement is unique: a-p---na_ becomes a-p---na-. The proposal replaces every underscore of the tag with a hyphen, so that in the tokens that also carry underscores in positions 2–8 — 1,316 in Gorman and one in Pedalion — it changes those positions as well and coincides with the proposal of underscore-person-number (S2).

S7.3 The positive written in the degree position

Gorman, same document, sentence 16, token 45: καλῶς, lemma καλός, tag d-------p. The ninth position contains p for the positive degree, but the scheme admits there only c for the comparative and s for the superlative: the positive is written with the hyphen. The check fires 1,218 times in Gorman and 454 in Pedalion. It is a systematic variant with automatic correction: the replacement is unique and touches no other position of the tag. The value is not arbitrary — it expresses the positive degree — and the difference is one of encoding, not of analysis.

S7.4 The part of speech b in Pedalion

Pedalion, Aristophanes, *Acharnians* (achar), sentence 1, token 9: δὲ, lemma δέ, tag b--------, relation COORD. The first position contains b, which does not belong to the AGDT inventory. It is not a value left there by mistake: the README of the Pedalion Trees declares it, "For experimental reasons, some trees exhibit the part-of-speech designation 'b', equalling in most cases 'd' or 'c'". The check fires 8,555 times in Pedalion, and 1,405 and 2,131 times in the two manual strata of the GLAUx samples: the presence of the same local value in those strata is compatible with those texts coming from the Pedalion tradition, but the datum alone does not prove it. The Pedalion Trees are among the treebanks from which the manually annotated part of GLAUx derives (GLAUx README), which makes that provenance plausible, although the README does not identify the source of individual texts. It is a systematic variant with manual review, and on its own it accounts for 51.6% of the variants of Pedalion. It is the only check of the catalogue whose authority is the project convention: it is refuted by showing that the declaration is not there.

S7.5 The common gender c

Pedalion, *Acharnians*, sentence 15, token 1: ἐγὼ, lemma ἐγώ, tag p1s---cn-. The seventh position contains c, a value that the AGDT 2.0 inventory does not list: in that inventory gender is only masculine, feminine or neuter. The check fires 8,256 times in Pedalion, 5,163 and 835 times in the two strata of the first GLAUx sample, 4,267 and 769 times in those of the confirmation sample, and never in Gorman. The distribution of the value is compatible with a local use for the common gender, but this interpretation is not documented in the available resources: none of the projects declares it. For this reason the check classifies it as a case to check with manual review and does not propose replacing c with m or with f, because an automatic replacement would decide in place of the reader. The value appears 8,257 times among the imported tokens of Pedalion and 8,256 times among the active tokens, because one of its occurrences sits on a technical token; the check, which looks at the active tokens with a nine-character morphological tag, reports 8,256 of them.

S7.6 Person on the personal pronoun

Gorman, *Against Timarchus*, sentence 1, token 77: ὑμῖν, lemma σύ, tag p-p---md-, relation OBJ_CO. The second position is empty, but ὑμῖν is a second-person personal pronoun. The AGDT 2.0 guidelines require gender, number and case for the pronoun, not person (Celano 2014, § 2.3, where the pronoun "is annotated for gender, number, and case"): the hyphen does not violate the scheme. This is the first of the two phenomena of § 6.3 of the article. The check fires 2,372 times in Gorman, 176 in Pedalion, 1,773 in the automatic stratum of the first sample of GLAUx and 1,451 in that of the confirmation sample, and only once in each of the two manual strata. It is a systematic variant with manual review: the annotation profile that GCP adopts as its reference fills in person on personal pronouns, and the corpus that leaves it empty regularly follows an annotation convention

different from the reference one. The check uses no thresholds: if the guidelines do not prescribe person, its absence is never a violation, whatever the rest of the corpus does. What decides is the closed list of the forms of the personal pronouns (S6), because the part of speech p is not enough: it also includes relatives, demonstratives and indefinites. The comparison with the list is exact — accents and capitalisation included — so that grave-accented or enclitic forms absent from it, such as σὺ or μοι, are not examined (S2).

S7.7 The relation label written bare: CO

GLAUx, first sample, Sophocles, *Oedipus at Colonus* (0011-007), sentence 8, token 12: ἐλαίας, lemma ἐλαία, tag n-s---fg-, relation CO. The AGDT scheme does not admit CO on its own: _CO is a suffix appended to the function of every conjunct of a coordination headed by COORD (e.g. OBJ_CO; Bamman and Crane 2008: 26–27; Celano 2014, §§ 3.9, 3.20.1), as in the analytical layer of the Prague Dependency Treebank (Hajič et al. 1999). In the automatic strata of the two GLAUx samples the bare label is systematic: it occurs 2,369 times among the imported tokens of the first sample, in all eighteen documents, and 1,887 times in the confirmation sample, in all seventeen; once the technical tokens are discarded, relation-bare-convention reports 2,239 and 1,782 findings, again in all eighteen and all seventeen documents. The catalogue classifies them as a case to check, for manual review and without a proposed correction: the distribution is that of a regular practice, but neither the README of GLAUx nor the guidelines file of its repository mentions CO, and diffusion alone does not turn a distribution into a declared convention.

In Gorman and in Pedalion bare CO is sporadic, as § 6.3 of the article states: 11 findings in 4 of the 158 documents of Gorman and 11 in 5 of the 40 of Pedalion. They are reported by relation-truncated-or-foreign, a verifiable violation for manual review, which in Gorman also reports 8 bare Aux and one CONJ, for 20 findings in 11 documents. The two diffusion thresholds of S4 are computed on the imported tokens, technical ones included: there bare CO occurs 14 times in 6 documents of Gorman, below both thresholds, and 32 times in 10 documents of Pedalion, below the threshold of diffusion. The same label thus changes class according to how widespread it is in the resource, and this is why the thresholds are declared. Where the systematic CO of GLAUx comes from cannot be established from the documentation: according to the GLAUx README, the treebanks used to train its automatic annotation include the Gorman and Pedalion Trees, in which bare CO is sporadic, while Keersmaekers et al. (2019: 111) describe, for the homogenised treebanks of the project, a conversion of coordination in which the coordinated nodes are attached to a preceding conjunct with a relation “CO”. The two phenomena of § 6.3 of the article are treated differently for reasons recorded in S3: for the pronoun the AGDT guidelines, which do not require person; for the bare label, when it is systematic, the internal usage of the resource.

S7.8 The decomposed encoding of GLAUx

GLAUx, first sample, *Oedipus at Colonus*, sentence 1, token 1: τέκνον, whose έ is stored as two characters, ε (U+03B5) followed by the combining acute accent (U+0301), instead of the single character έ (U+03AD). GLAUx is published in this decomposed form, NFD, as its README declares: “the GLAUx corpus is encoded in Unicode-NFD, meaning that diacritics are separate characters”. In the automatic stratum of the first sample the form or the lemma is not in NFC in 252,406 of the 292,567 active tokens, 86.3% (§ 6.4 of the article): in 245,175 because the diacritics are decomposed; in the other 7,231 because the token contains a character that Unicode

normalisation replaces in either form — the ano teleia U+0387 (6,845 tokens, nearly all of them the punctuation mark itself) or the keraia U+0374 (383) — or, in three lemmas, a mixture of composed and decomposed characters or combining marks in a non-canonical order. Since the share exceeds the threshold of 50% (S4), corpus-not-nfc emits a single finding for the whole resource, on the first active token not in NFC, with the count in its parameters: it is the only check that reports on the corpus as a whole. In Pedalion, by contrast, 4,007 of the 274,609 active tokens (1.5%) are not in NFC, in a resource that is otherwise in NFC: there unicode-not-nfc reports them token by token, and most of them are not decomposed but written with characters that NFC replaces with their canonical equivalents, such as the ano teleia U+0387 and the vowels with oxia. Both checks count the active tokens, so that for Pedalion the denominator is not the 281,245 imported tokens. 86.3% of one resource is how that resource is published; 1.5% of another is a minority encoding within it.

S7.9 δεῖν annotated as a participle, and the two passages that the lexicon marks as doubtful

The form δεῖν is the infinitive of δεῖ; the corresponding participle is δέον (LSJ s.v. δεῖ; Smyth § 397 a). In Gorman the form occurs 165 times among the words, and 37 occurrences carry p in the mood position. The catalogue treats them with two twin checks. The first fires 36 times and proposes the automatic correction, because the infinitive is the only possible reading: it is a verifiable violation. The second fires only once, on the occurrence that falls in Lysias 14, and proposes nothing: LSJ, s.v. δεῖ, IV, marks as doubtful two attestations of participial δεῖν, Lysias 14.7 and Aristophanes, fr. 220, and that occurrence belongs to the speech that contains the first of them. It is a case to check with manual review. In the catalogue the two checks remain distinct and are counted separately: dein-participle with 36 findings and dein-participle-disputed with one. In Table S2.1 they are presented together as a single diagnostic procedure on the phenomenon δεῖν, which comprises 37 occurrences annotated as a participle out of 165 occurrences of the form among the words. The list of the two passages is closed and declares its own source (S6); it also declares at what granularity each passage is identifiable, because a token whose provenance fields give neither a CTS citation nor both the author and the work numbers does not make it possible to exclude either of the two, and its occurrence goes to manual review in any case (S2).

S7.10 The two encodings of the apostrophe

Pedalion, *Acharnians*, sentence 3: “φέρ’ ἴδω, τί δ’ ἥσθην ἄξιον χαιρηδόνος;”. In the same file 637 elision apostrophes are encoded as U+2019 and 18 as U+02BC. Neither of the two characters is wrong in itself, and the check does not say which one is right: it reports that two code points for the same sign coexist within the same file, which makes two strings that are identical on screen different. The check fires 2,394 times in Pedalion and 1,655 in Gorman, and the situation occurs in 11 of the 40 files of Pedalion and in 16 of the 158 of Gorman. It is a systematic variant with automatic correction, because the conversion to a single canonical character is mechanical.

S8. Variation nuclei, by level of analysis

The variation n-gram method (Dickinson and Meurers 2003a), extended to dependencies by Boyd, Dickinson and Meurers (2008), looks within a corpus for strings that occur more than once with different annotations. Such a string is a variation nucleus — a single word for part-of-speech annotation, a pair of words for dependencies — and the nucleus together with the identical words that surround it is a variation n-gram (Boyd, Dickinson and Meurers 2008: 115–116). GCP uses an adaptation of the method, on the five levels described below (§ 6.1 of the article), and finds 33,652

nuclei in Gorman, 13,848 in Pedalion and 11,347 in the automatic stratum of the first GLAUx sample (Table S8.1). The method uses none of the 77 checks and no knowledge of Greek, and this is why it serves as an external criterion.

The five levels are the following; for each, the nucleus is named first and then the annotation that varies. Morphology: a form, which receives different morphological tags. Tag positions: a form, which receives different values in one position of the nine-character tag, the hyphen not counting as a value; each position is considered separately, so that a form that varies in two positions counts as two nuclei. Context: a form preceded and followed by the same forms, which receives different morphological tags; a form that varies in two different contexts counts as two nuclei. Dependencies: a pair of forms linked by a dependency, which receives different labels, the label being the base relation (S2) together with the side of the head, left or right of the dependent. Lemmas: a form with its part of speech, which receives different lemmas.

All five levels work on the words (S11.1), with forms and lemmas compared in NFC, and share one parameter in all strata: a nucleus is counted if it occurs at least twice. Only the context level uses the surrounding words, one on each side — the minimal context of the non-fringe heuristic (Dickinson and Meurers 2005: 328), which Boyd, Dickinson and Meurers (2008: 116) report to be sufficient for detecting errors with high precision; the other four levels compare the nucleus without context. The morphology and context levels apply the method of Dickinson and Meurers (2003a) to the morphological tag. The dependency level adopts the representation by pairs of Boyd, Dickinson and Meurers (2008: 123), in which the nucleus is a pair of words linked by a dependency and the label records the direction of the head; it does not implement the other components of their method — the NIL nuclei, which compare a linked pair with an occurrence of the same two words not linked to each other (2008: 123–124), the treatment of overlapping type-identical dependencies as a single set of labels (124–125) and the three context heuristics (126–128) — so that its nuclei are not filtered by context. The tag-position and lemma levels are additions of GCP. No threshold has been adapted to a corpus.

Table S8.1 Variation nuclei by level of analysis and by stratum. The total of each row is the sum of the five levels; a point of the text may appear on more than one level, because the levels look at different objects.

Stratum	Morphology	Positions	Context	Dependencies	Lemmas	Total
Gorman	6,159	8,259	3,526	13,416	2,292	33,652
Pedalion	3,427	4,534	735	4,432	720	13,848
GLAUx automatic	2,956	3,600	586	4,158	47	11,347
GLAUx manual	382	495	3	98	18	996
GLAUx confirmation automatic	2,554	3,284	580	4,623	10	11,051
GLAUx confirmation manual	297	345	22	372	5	1,041

S8.1 The concentration, and how it is verified

§ 6.1 of the article compares the points identified by the variation n-gram with those reported by the program. The unit of the comparison is the distinct form, not the occurrence: the question is whether the forms involved in a variation are reported more often than the forms of the corpus taken all together. The forms are counted among the words and compared in composed form (NFC), so that

two spellings identical on screen are not counted as distinct: in GLAUx, which publishes in NFD, and in Pedalion, where part of the forms are written with characters that NFC replaces with their canonical equivalents, such as the vowels with oxia (S7.8); for this reason the number of distinct forms of Pedalion used here, 45,132, is lower than the 45,410 of Table 2 of the article, which counts the strings as they stand in the files. A form is "involved in a variation" if it appears as a nucleus on at least one of the five levels; on the dependency level the nucleus is a pair, and both forms count.

Table S8.2 The contingency table of the comparison, by stratum. N = distinct forms among the words, in NFC; K = forms of N that coincide with the form of at least one token carrying a finding; n = forms involved in at least one variation; k = forms that are both. "Expected" is the number of reported forms that one would expect within n if the two properties were independent, that is, n·K/N. In the manual stratum of the first GLAUx sample one of the 240 forms of K comes from a finding on a punctuation token whose string also occurs as a word; counting only the findings on words would give K = 239.

Stratum	N	K	n	k	Expected	Share in the nucleus	Share in the corpus	Ratio
Gorman	69,329	5,928	12,021	2,973	1,028	24.7%	8.6%	2.89
Pedalion	45,132	4,186	5,593	1,451	519	25.9%	9.3%	2.80
GLAUx automatic	48,502	2,999	5,074	1,091	314	21.5%	6.2%	3.48
GLAUx manual	7,219	240	461	70	15	15.2%	3.3%	4.57
GLAUx confirmation automatic	38,234	2,453	4,580	999	294	21.8%	6.4%	3.40
GLAUx confirmation manual	7,159	252	587	65	21	11.1%	3.5%	3.15

The test, reported stratum by stratum in Table S8.3, is hypergeometric and directional: the null hypothesis is that the k forms are distributed as in a draw without replacement of n forms from an urn of N forms, K of which are reported, and the alternative hypothesis is that within the nucleus the reported forms are in excess. The value reported is therefore the upper tail, that is, the probability of observing at least k reported forms under the null hypothesis. There is a single family of comparisons, and it comprises the six strata; the correction is that of Holm (1979), chosen instead of Bonferroni's because it controls the same family-wise error rate and is uniformly less conservative.

Table S8.3 The test, stratum by stratum. The ratio is the share in the nucleus divided by the share in the corpus. The p-values of Gorman, Pedalion and the automatic stratum of the first GLAUx sample are below the smallest positive number representable in double precision (about $4.9 \cdot 10^{-324}$): they are given as computed in logarithmic space from the counts of Table S8.2. The p-value of the automatic stratum of the confirmation sample, $9.9 \cdot 10^{-314}$, is still representable, as a subnormal number.

Stratum	Ratio	Nominal p	Corrected p (Holm)	Significant at 0.05
Gorman	2.89	$6.6 \cdot 10^{-820}$	$4.0 \cdot 10^{-819}$	yes
Pedalion	2.80	$3.7 \cdot 10^{-347}$	$1.9 \cdot 10^{-346}$	yes
GLAUx automatic	3.48	$3.7 \cdot 10^{-342}$	$1.5 \cdot 10^{-341}$	yes
GLAUx manual	4.57	$4.4 \cdot 10^{-29}$	$8.8 \cdot 10^{-29}$	yes
GLAUx confirmation automatic	3.40	$9.9 \cdot 10^{-314}$	$3.0 \cdot 10^{-313}$	yes
GLAUx confirmation manual	3.15	$9.6 \cdot 10^{-18}$	$9.6 \cdot 10^{-18}$	yes

The concentration therefore remains significant in all six strata after the correction. What this means must be stated precisely: the test measures an association between two ways of selecting the same forms, not a causal relation and not the correctness of the annotations. The two criteria are built independently of each other, but this does not make them statistically independent: both may depend on properties of the forms, such as their frequency, their ambiguity and the number of occasions on which they can vary, and no test stratified by such properties has been carried out.

Neither of the two criteria proves that the forms concerned are badly annotated, and concentration at the level of the form does not imply that the individual occurrences identified are the same.

The count can be redone from the files of the runs (S11.4): the five files variation_*.parquet give the nuclei with the column nucleo, findings.parquet gives the reported forms and tokens.parquet gives the words. The four figures N, K, n and k of Table S8.2 were recalculated from those files for all six strata.

S8.2 What the lemma level shows

The lemma level sharply separates the manually revised corpora from the automatic strata: 2,292 nuclei in Gorman and 720 in Pedalion against 47 and 10 in the two automatic strata of GLAUx. By the criteria adopted, little variation of the lemma is observed in the automatic strata; this measures the consistency of those samples, not the correctness of the lemmas, and it is not a guarantee that follows from automatic annotation: a lemmatiser can assign different lemmas to the same form in different contexts, as the nuclei that remain show.

S9. The two GLAUx samples

GLAUx (Keersmaekers 2021) contains about twenty million tokens of Ancient Greek in XML files whose annotation, mostly automatic, follows the AGDT guidelines “roughly”, in the words of its README. The two samples of the study were drawn from the same pool of eligible texts with two seeds, the second from the texts left by the first. The procedure is given below; the forty texts are listed in Table S9.1 (§ 6.2 of the article).

The procedure has five steps. (1) The pool consists of the texts of the GLAUx metadata table distributed with the program (data/glaux_metadata.csv) whose TLG author number is not that of an author of Gorman or Pedalion and whose size declared in the metadata is between 5,000 and 60,000 tokens: 338 texts. The table contains 1,421 rows, one per text, all with a TLG author number and a TLG work number; in 23 rows the work number is followed by a letter that distinguishes parts of the same work (for instance 0096-002a and 0096-002b), and these rows enter the pool like the others. The author number of a text is the first four digits of its identifier, document_id (0096 in 0096-002a), which in every row coincide with the column tlg_author; the test of the program on the absence of overlap compares the same four digits. (2) The authors of Gorman and Pedalion are identified by the TLG author numbers written in their XML files, 37 numbers, which are also listed in the test of the program on the absence of overlap (tests/test_pipeline_base.py): 0003, 0006, 0007, 0008, 0009, 0010, 0014, 0016, 0019, 0026, 0028, 0032, 0041, 0058, 0059, 0060, 0062, 0081, 0086, 0093, 0096, 0343, 0526, 0537, 0540, 0541, 0543, 0544, 0551, 0554, 0557, 0559, 0561, 0585, 1220, 2003 and 4029. (3) The first sample is drawn from the pool, taken in the order of the metadata table, with random.seed(20260814) followed by random.sample of twenty texts, from the standard library of Python. (4) The confirmation sample is drawn with random.seed(20260815) from the 318 texts that remain, sorted by TLG identifier. (5) Within each sample the texts annotated by hand form a stratum of their own, because their annotation has a different origin: two texts in the first sample and three in the confirmation sample.

The size of a text is the number of tokens declared in the GLAUx metadata, which counts different units from the tokens imported by the program; counted in imported tokens, the forty texts range from 5,109 to 41,942 and all fall within the same interval. The metadata tables of Gorman and Pedalion distributed with the program give 42 author numbers instead of 37, because they assign

one also to texts whose XML carries none; used in place of the 37, they give a pool of 299 texts and do not reproduce the draw. The draw is not performed by the program, which reads its result, the list of the forty texts in data/glaux_sample.csv; the same file is the list of files that the program downloads, so that the published list and the one used cannot diverge. With the metadata table and the 37 author numbers, both distributed with the program, the procedure reproduces the two draws exactly, twenty texts out of twenty in both samples, under Python 3.13 and 3.14; across versions of Python only the sequence of random() is guaranteed to be the same for the same seed, not that of sample() (Python Software Foundation n.d.).

The two samples have different roles (§ 6.2 of the article). The first, the development sample, contributed to the construction of the catalogue and is therefore not an independent validation: the criteria of four checks — corpus-not-nfc, lemma-graphic-variant, relation-bare-convention and uninflected-lemma — were formulated on its automatic stratum (§ 4.4 of the article), and the source code of these checks gives the figures of that stratum on which they rest, such as the 2,369 bare CO in all eighteen documents (syntax.py). The confirmation sample was drawn with a different seed from the texts that the first had left. It is a second sample, not a test set kept apart: it too was analysed while the catalogue was still being revised. What it shows is that the automatic stratum of a separate draw gives a profile similar to that of the first (Table 6 of the article), not how the criteria behave on texts never examined.

No text of the two samples shares a work with Gorman or Pedalion (§ 6.2 of the article): neither by TLG work identifier nor, for the twelve files of Pedalion that carry none, by the work identified by the file. The exclusion by author relies on the TLG numbers written in the XML files, and ten of the forty files of Pedalion carry none: example-sentences, the three files of Genesis, mimn, paean, papyri, pseudo-lucian_themule, semonides and theoc. Their authors or collections — the Septuagint, Theocritus, Pseudo-Lucian, Mimnermus, Semonides and Paeanius — are therefore not among the 37 numbers excluded, and one text of the first sample, 0527-012 (Septuaginta, Regnorum ii), belongs to the same TLG corpus, 0527, as the Genesis files of Pedalion. GLAUx, moreover, is not independent of the two corpora as a resource: according to its README, the treebanks that it incorporates, the Gorman and Pedalion Trees among them, “have also been used as training material for the automatic annotation”.

At the level of the passage the samples share little text with the two corpora. Comparing the words of each text with those of Gorman and Pedalion in sequences of at least five consecutive words of the same sentence — forms without diacritics, in lower case and with the elision marks unified — 37 of the 40 texts share at least one sequence, for 3,861 of their 525,904 words (0.7%). In 26 texts the shared words are fewer than 100 and are mostly formulaic sequences; in 11 they are 100 or more and come from quotations: Porphyry, *De abstinentia*, quotes Plutarch’s *Lycurgus* (598 shared words, 483 of them with that life); the rhetorical treatises of Aelius Aristides, Hermogenes and Apsines quote Demosthenes, Thucydides and Xenophon (598, 546 and 183); Philo quotes Genesis in two treatises (244 and 145); Lycurgus, *Against Leocrates*, shares 194 words with the orators and historians of Gorman; the example sentences of Pedalion include lines of Sophocles, *Oedipus tyrannus*, and of Aeschylus, *Agamemnon* (153 and 121); lines of Hesiod, *Works and Days*, are quoted in Xenophon’s *Memorabilia* in Pedalion and by Aristotle and Aeschines in Gorman (100 words in all); and Regnorum ii shares 129 words with the Genesis of Pedalion. The shared words are at most 8.7% of the words of a text (Hermogenes) and less than 3% in all the others.

Table S9.1 The forty texts of the two samples. The column “Tokens” gives the number declared by the metadata of the resource, which does not coincide with the tokens imported by the program, because it counts different units. The year is the initial one declared by the metadata. The last column gives the first twelve hexadecimal characters of the SHA-256 of the XML file read by the corresponding run, recomputed on the files of the download folder of the runs (S11.4).

TLG	Sample	Stratum	Author	Work	Genre	Dialect	Year	Tokens	SHA-256 (first 12)
0284-042	first	automatic	Aelius Aristides	Περὶ ὁμονοίας ταῖς πόλεσιν	Oratory	Attic/Koine	101	7,257	81dd4de9b2e6
0082-003	first	automatic	Apollonius Dyscolus	De conjunctionibus	Language	—	101	14,218	17a60450f7c0
0552-004	first	automatic	Archimedes	De lineis spiralibus	Mathematics	Doric	-300	15,871	0ac0ee2c3008
0556-001	first	automatic	Asclepiodotus	Tactica	Military	—	-100	7,414	10a3dacf990d
0029-004	first	automatic	Dinarchus	In Demosthenem	Oratory	—	-400	8,577	977a937f4fa4
1799-012	first	automatic	Euclides	Phaenomena	Astronomy/Astrology	—	-300	9,197	5e44c3b0309a
0057-049	first	automatic	Galenus	Adversus eos qui de typis scripserunt vel de circuitibus	Medicine	Attic/Koine	101	6,298	a2c16d942062
0627-010	first	automatic	Hippocrates et Corpus Hippocraticum	De articulis	Medicine	Ionic	-500	25,227	f2338e56868e
1386-001	first	automatic	Historia Alexandri Magni	Recensio α sive Recensio vetusta	Narrative	—	201	38,750	97e1b14011f4
2042-045	first	automatic	Origenes	Epistula ad Africanum	Epistolography	Koine	101	6,223	8ebf25d1b496
2042-075	first	automatic	Origenes	Expositio in Proverbia	Commentary	Koine	101	21,229	bf84c7fc2b0d
0018-031	first	automatic	Philo Judaeus	Legatio ad Gaium	Philosophy	Koine	-100	20,091	6ba999df1434
1600-001	first	automatic	Philostratus Major	Imagines	Art	—	101	26,321	a4d227bfca9e
1608-001	first	automatic	Phrynichus	Praeparatio sophistica	Language	Attic/Koine	101	17,197	b9bfdf7313c5
0527-012	first	automatic	Septuaginta	Regnorum ii	Religious History	Koine	-200	20,238	093f1f8fb5c0
0011-007	first	automatic	Sophocles	Oedipus Coloneus	Tragedy	Attic	-500	12,802	d04639c56332
1766-001	first	automatic	Tatianus	Oratio ad Graecos	Oratory	—	101	11,829	9549c3818f7a
0098-001	first	automatic	Zenobius Sophista	Epitome collectionum Lucilli Tarrhaei et Didymi	Language	—	101	24,313	838059cd3ab9
0085-005	first	manual	Aeschylus	Agamemnon	Tragedy	Attic	-600	9,760	ea15665b808e
0011-004	first	manual	Sophocles	Oedipus tyrannus	Tragedy	Attic	-500	11,374	ef18605abc80
0284-056	confirmation	automatic	Aelius Aristides	Ars rhetorica	Rhetoric	Attic/Koine	101	27,127	da5e9c9d7ac6
0284-027	confirmation	automatic	Aelius Aristides	Ἱεροὶ λόγοι ε΄	Biography	Attic/Koine	101	5,052	60f4f23fcbaf
0732-011	confirmation	automatic	Alexander	De anima libri mantissa	Philosophy	—	101	40,753	16aff3635db2
0057-046	confirmation	automatic	Galenus	De morborum temporibus liber	Medicine	Attic/Koine	101	6,087	c28489a93409
1419-001	confirmation	automatic	Hermas	Pastor	Religious Prophecy	Koine	101	32,014	82b09a6c184a
0592-005	confirmation	automatic	Hermogenes	Περὶ μεθόδου δεινότητος	Rhetoric	Attic/Koine	101	7,599	99e6562f6c99
0627-025	confirmation	automatic	Hippocrates et Corpus Hippocraticum	De affectionibus	Medicine	Ionic	-500	8,734	d21b5f8f9f52
0627-003	confirmation	automatic	Hippocrates et Corpus Hippocraticum	Prognosticon	Medicine	Ionic	-500	6,087	1a1fa54ed8fb
0034-001	confirmation	automatic	Lycurgus	Oratio in Leocratem	Oratory	Attic	-400	12,421	6cbd1e8421e6
0018-018	confirmation	automatic	Philo Judaeus	De mutatione nominum	Theology	Koine	-100	15,358	e8ea99e655bd
0018-006	confirmation	automatic	Philo Judaeus	De posteritate Caini	Theology	Koine	-100	12,633	db12c6a6b4da
0018-026	confirmation	automatic	Philo Judaeus	De praemiis et poenis et De exsecrationibus	Philosophy	Koine	-100	10,323	0c45c5a44064
1608-002	confirmation	automatic	Phrynichus	Eclogae	Language	Attic/Koine	101	10,322	04333b57b4c1

TLG	Sample	Stratum	Author	Work	Genre	Dialect	Year	Tokens	SHA-256 (first 12)
2034-003	confirmation	automatic	Porphyrius	De abstinentia	Philosophy	—	201	39,825	46dd224f7c52
2034-002	confirmation	automatic	Porphyrius	Vita Pythagorae	Biography	—	201	6,551	01907188e930
2027-001	confirmation	automatic	Valerius Apsines	Ars rhetorica	Rhetoric	—	201	21,927	130dda4bf2a7
1747-001	confirmation	automatic	Vitae Adam Et Evae	Vita Adam et Evae	Religious Narrative	Koine	1	5,129	c921c18bbe11
0020-002	confirmation	manual	Hesiodus	Opera et dies	Epic poetry	Ionic/Epic	-800	6,954	7f1bb9f1b2df
0031-006	confirmation	manual	Novum Testamentum	Epistula Pauli ad Romanos	Religious Epistle	Koine	1	8,191	828fb245e8dd
0031-002	confirmation	manual	Novum Testamentum	Evangelium secundum Marcum	Religious History	Koine	1	12,998	5f431174c49c

Counting the tokens actually imported by the program, the first sample gives 300,292 tokens for the eighteen automatic texts and 21,652 for the two manual ones; the confirmation sample gives 273,354 for the seventeen automatic texts and 28,666 for the three manual ones. In all, 623,964 imported tokens out of about twenty million: enough to verify the behaviour of the checks on different data, not to describe GLAUx as a whole.

One text weighs on the rate of violations of the automatic stratum of the first sample (§ 6.4 of the article). 0527-012 (Septuaginta, *Regnorum ii*), with 20,238 active tokens, produces 253 of the 378 verifiable violations of the stratum, all from slot-missing, mostly on indeclinable proper names of Semitic origin, such as Ἀμαλήκ, Χεβρών and Νάβαλ, whose gender or case is left unfilled. The stratum has 378 violations on 292,567 active tokens, 12.9 per 10,000; without that text it has 125 on 272,329, 4.6 per 10,000.

S10. The comparison with two automatic analysers

This section documents the comparison of § 7 of the article: what is compared and with which models (S10.1), the denominators (S10.2), the definitions of the measures (S10.3), the measures by field, by position of the tag and by part of speech (S10.4–S10.7), the three-way comparison (S10.8), the recurrent substitutions (S10.9), the lemma (S10.10), what is left out (S10.11), how the predictions were obtained and how the figures are recomputed (S10.12–S10.13), and the overlap between the evaluated corpora and the training data of the models (S10.14).

S10.1 Provenance of the four comparisons

What is compared. Both models are trained on UD_Ancient_Greek-Perseus, the Universal Dependencies conversion of passages of AGDT 2.1, whose XPOS column keeps the nine-character AGDT morphological tag (UD_Ancient_Greek-Perseus README): all the 159,895 tokens of its training set carry such a tag in releases 2.4 and 2.18, and both models write it in the XPOS column of their output. The comparison therefore concerns the lemma and the XPOS tag, in the format of the corpora; UD part-of-speech tags (UPOS), morphological features and dependency relations are not compared (S10.11). The models receive the active tokens of the corpus — in Pedalion all but one, whose form is empty (Table S10.2) — in the same order and divided into the same sentences, but none of its annotations (S10.12).

The models. Stanza (Qi et al. 2020), version 1.14.0, was run with its default Ancient Greek package, “perseus” (tokenize: perseus; pos and lemma: perseus_nocharlm; pretrained vectors: conll17), with the processors tokenize, pos and lemma and with pretokenised input (tokenize_pretokenized=True),

so that it keeps the tokenisation and the sentence division of the corpus; GCP reads the predicted XPOS and lemma. UDPipe 1 (Straka, Hajič and Straková 2016) was run through the Python binding ufal.udpipe 1.4.0.1 with the model ancient_greek-perseus-ud-2.4-190531.udpipe (Straka and Straková 2019), trained on Universal Dependencies 2.4; for this model the documentation of UDPipe 1 reports, on the UD Perseus test set and from raw text, an accuracy of 72.2% on XPOS and of 82.7% on the lemma (UDPipe 1 models documentation, UD 2.4 models). The release of UD on which the Stanza models were trained is not documented in the materials of the runs (S10.14).

Table S10.1 Provenance of the four comparisons. The two fingerprints are defined below the table; the column "Start" gives the UTC time at which each comparison phase began.

Corpus	Model	Version	Corpus fingerprint	Resource fingerprint	Tokens supplied to the model	Start (UTC)
Gorman	Stanza	1.14.0	d7f08464469afa39	738ee3df02f33fa5	616,646	2026-09-09 05:36
Gorman	UDPipe	1.4.0.1	d7f08464469afa39	28e919b3269a6730	616,646	2026-09-09 05:47
Pedalion	Stanza	1.14.0	2e3f8845e9df6259	738ee3df02f33fa5	274,608	2026-09-09 06:09
Pedalion	UDPipe	1.4.0.1	2e3f8845e9df6259	28e919b3269a6730	274,608	2026-09-09 06:38

The corpus fingerprint (sixteen hexadecimal characters) identifies the text supplied to the model: it is the SHA-256 of the sequence, in sentence and token order, of the sentence key and the NFC form of the tokens counted in the penultimate column. It changes if a token, their order or the filter that decides which tokens pass changes; since it is computed on the NFC form, while the model receives the form as it stands in the file, it identifies that text up to Unicode normalisation (in Pedalion 3,586 forms supplied are not in NFC). The resource fingerprint, also of sixteen characters, is computed on the relative path and the size of every file in the folder of the model, without reading their content: it shows that the files had the same names and sizes, not that their bytes were identical. For UDPipe 1.4.0.1 it covers the model file ancient_greek-perseus-ud-2.4-190531.udpipe (20,492,125 bytes); for Stanza 1.14.0 it covers resources.json and the four files that the default grc package uses for the three processors: tokenize/perseus.pt, pos/perseus_nocharlm.pt, lemma/perseus_nocharlm.pt and pretrain/conll17.pt. The run does not record the list of files, but both resource fingerprints, recomputed on these files in the model folder of the program, give the values of the table; Table S11.3 gives the SHA-256 of each file, which identifies its content. Neither fingerprint is the twelve-character findings fingerprint of S11.3. The filter is the configuration adopted in the study (§ 3.1 of the article): technical tokens are excluded, while punctuation and non-Greek forms remain among the active tokens.

The four comparisons — Stanza and UDPipe on Gorman, Stanza and UDPipe on Pedalion — are the two comparison phases, one per model, of the runs 20260909-053400-gorman and 20260909-060742-pedalion, carried out on 9 September 2026 on a Windows machine with GCP 3.1.0 (Tables S11.1 and S11.2); they are not separate runs, and the diagnosis and the comparison come from the same imported tokens. The measures of this section are computed on the predictions saved by those phases. The corpus fingerprint and the number of tokens supplied to the models were recomputed and match Table S10.1.

S10.2 The denominators, one by one

Table S10.2 distinguishes the main sets used as denominators in the comparison. Not all the rows form a chain of subtractions: imported tokens, active tokens, tokens supplied to the models and words are sets defined by different criteria, while the last two rows are subsets of the words. Every

other denominator used in this section is a subset of one of these, declared at the point where it is used.

Table S10.2 The denominators of the comparison. The units are those defined in § 3.1 of the article.

Unit	Gorman	Pedalion	How it is defined
imported tokens	625,176	281,245	all the XML elements imported by the program according to the definition in § 3.1
active tokens	616,646	274,609	imported tokens after the exclusion of the technical tokens according to the adopted filter configuration; punctuation remains among the active tokens
tokens supplied to the models	616,646	274,608	active tokens; in Pedalion one active token with an empty form is also excluded
words	548,912	238,413	imported tokens that are neither technical nor punctuation, according to the definition in § 3.1
words with a reference lemma	548,803	238,389	words minus 109 and 24 cases in which the lemma field is empty
words with a nine-character tag	548,593	238,311	words minus 22 and 12 cases without a tag and 297 and 90 cases with a tag whose length is not nine

S10.3 The definitions of the measures

Agreement on the lemma. Numerator: the words in which the lemma key of the corpus is equal to the key of the predicted lemma. Denominator: the words with a reference lemma. The key is the composed form (NFC) stripped of the Perseus lemma numbering, applied to both sides: λέγω3 in the corpus and λέγω in the model are the same lexical choice. The map of canonical variants is not applied.

Agreement on the tag. Numerator: the words in which the tag string of the corpus is identical to the predicted one and the prediction is interpretable. Denominator: the words with a nine-character tag. A prediction is interpretable if it has nine characters and every position contains either the hyphen or one of the values of the AGDT inventory as the program encodes it, which is that of Table 1 of the article with, in addition, c in the position of gender; no prediction of the four comparisons contains c, and the two inventories give the same result. A non-interpretable prediction remains in the denominator and counts as a disagreement, even when it coincides with the tag of the corpus; the number of such predictions is reported separately, and it is never a threshold.

Coverage. The share of eligible words for which the model produced a prediction. It is 100.0% in all four comparisons: no measure in this section is reduced by missing predictions.

Agreement in one position of the tag. Numerator: the words in which the character of the corpus in that position is equal to the predicted one and the predicted character is interpretable in that position. Denominator: the eligible words in which the corpus fills the position, that is, in which its

character is not the hyphen. The denominator therefore also includes values of the corpus outside the inventory, such as the underscore or the p of the position of degree, which no model writes: S10.6 shows what this implies for the ninth position.

The three-way comparison. Corpus, Stanza and UDPipe on the same word, among the words eligible for the field. Three relations are evaluated: corpus and Stanza agree, and corpus and UDPipe agree, according to the pairwise definitions given above — for the tag, a non-interpretable prediction never agrees with the corpus; Stanza and UDPipe agree when their predictions coincide, for the lemma after the same normalisation and for the tag only if both predictions are interpretable (Stanza produces no prediction that is not interpretable, so that this condition changes no count). The five outcomes are defined by these three relations: all three agree; only corpus and Stanza agree; only corpus and UDPipe agree; only Stanza and UDPipe agree; no pair agrees. The last outcome therefore also includes the two Gorman words in which UDPipe writes the same tag as the corpus, a tag that is not interpretable (S10.5). The alignment between the two comparison tables uses the quadruple of source file, document, sentence index and token index, which identifies each word in both corpora.

S10.4 Agreement on the lemma and on the tag

Table S10.3 The eight main measures, each as numerator, denominator and rate.

Corpus	Model	Field	Agreements	Denominator	Rate	Interpretable predictions	Malformed	Coverage
Gorman	Stanza	lemma	489,727	548,803	89.24%	—	—	100.0%
Gorman	UDPipe	lemma	447,567	548,803	81.55%	—	—	100.0%
Pedalion	Stanza	lemma	204,950	238,389	85.97%	—	—	100.0%
Pedalion	UDPipe	lemma	187,521	238,389	78.66%	—	—	100.0%
Gorman	Stanza	tag	456,724	548,593	83.25%	548,593 / 548,593	0	100.0%
Gorman	UDPipe	tag	366,471	548,593	66.80%	548,590 / 548,593	3	100.0%
Pedalion	Stanza	tag	170,049	238,311	71.36%	238,311 / 238,311	0	100.0%
Pedalion	UDPipe	tag	138,585	238,311	58.15%	238,304 / 238,311	7	100.0%

To one decimal place, as in Table 9 of the article, the eight rates are 89.2% · 81.6% · 86.0% · 78.7% · 83.3% · 66.8% · 71.4% · 58.2%.

S10.5 The malformed predictions, one by one

UDPipe produces ten malformed predictions out of the 786,904 words eligible for the tag in the two corpora, that is, 548,593 in Gorman and 238,311 in Pedalion; Stanza produces none. The program lists them instead of merely counting them, because a prediction that cannot be read has to be looked at before it is classified.

Table S10.4 The predictions of UDPipe that do not have the AGDT positional form. The value g in the position of mood and the value d in the position of voice do not belong to the AGDT inventory for Greek (Table 1 of the article): they are values that the positional tagset of the same family of treebanks uses for Latin. SYM is not a positional string.

Corpus	Prediction	Occurrences	Why it is not interpretable	The corpus tag at the same point
Gorman	v-sagamn-	2	mood g, outside the inventory	v-sapamn- (ἐξελκύσας) and v-sagamn- (ἀκούσας)

Corpus	Prediction	Occurrences	Why it is not interpretable	The corpus tag at the same point
Gorman	v--pnd---	1	voice d, outside the inventory	v--pnd--- (συνέχεσθαι)
Pedalion	SYM	6	not a nine-character string	n-_---__- three times, n--------, n-s---nn- and a-------- (the form is ∠ in all six)
Pedalion	v-sagamn-	1	mood g, outside the inventory	v-sapamn- (ἀκούσας)

In Gorman, in two of the three occurrences the prediction coincides letter for letter with the tag written in the corpus, which therefore itself carries a value outside the inventory. The prediction counts as a disagreement because it cannot be read in the scheme, not because it differs from the corpus: this is the intended consequence of the definition given in S10.3, and the difference it makes to the total for Gorman and UDPipe is two words out of 548,593.

S10.6 Position by position

Table S10.5 Agreement position by position, with the explicit denominator. The rates of different rows cannot be compared with one another as if they had the same base: each position concerns a different subset of the lexicon.

Pos.	Feature	Gorman: words	Stanza	UDPipe	Pedalion: words	Stanza	UDPipe
1	part of speech	548,583	503,965 = 91.9%	429,540 = 78.3%	238,298	197,668 = 82.9%	173,578 = 72.8%
2	person	54,241	49,335 = 91.0%	46,596 = 85.9%	35,794	27,790 = 77.6%	25,653 = 71.7%
3	number	366,966	359,667 = 98.0%	351,260 = 95.7%	158,248	153,776 = 97.2%	149,400 = 94.4%
4	tense	107,642	101,562 = 94.4%	88,976 = 82.7%	50,508	46,463 = 92.0%	39,957 = 79.1%
5	mood	107,647	104,264 = 96.9%	98,992 = 92.0%	50,555	47,809 = 94.6%	44,490 = 88.0%
6	voice	107,610	102,868 = 95.6%	95,985 = 89.2%	50,503	47,688 = 94.4%	44,306 = 87.7%
7	gender	316,327	291,854 = 92.3%	278,582 = 88.1%	128,610	108,957 = 84.7%	101,791 = 79.1%
8	case	316,706	302,321 = 95.5%	289,842 = 91.5%	128,933	119,247 = 92.5%	112,121 = 87.0%
9	degree	16,476	2,993 = 18.2%	1,506 = 9.1%	3,354	1,319 = 39.3%	619 = 18.5%

Degree. The models rarely fill the ninth position, and only with c or s: Stanza does so on 3,516 words of Gorman and UDPipe on 2,271, out of 548,593, while the corpus fills the position on 16,476 words — 11,187 times with the underscore and 1,218 times with the p of the positive, the values reported by degree-underscore and grade-p, and only 4,071 times with the comparative c or the superlative s. On the 12,405 words with the underscore or p agreement is zero by construction, because no model writes these values; on the 4,071 words with c or s Stanza agrees 2,993 times (73.5%) and UDPipe 1,506 times (37.0%). In Pedalion the corpus fills the position on 3,354 words (270 underscores, 454 p and 2,630 c or s), Stanza on 1,483 and UDPipe on 996; on the words with c or s Stanza agrees 1,319 times (50.2%) and UDPipe 619 times (23.5%). In Gorman, therefore, the rates of Table S10.5 for degree register mostly the conventions with which the corpus writes the position (12,405 of 16,476 words). In Pedalion these conventions are 724 of 3,354 words, and most

disagreements fall on comparatives and superlatives on which the model leaves the position empty: 1,311 of the 2,035 disagreements of Stanza and 2,011 of the 2,735 of UDPipe.

Person. § 7.2 of the article reports that agreement on the second position is lowest in the combination Pedalion × UDPipe, 71.7%, and that it must be read with caution for pronouns, on which AGDT does not require person. The figures are these. Out of 35,794 words in which Pedalion marks person, UDPipe agrees 25,653 times and diverges 10,141 times; in 8,086 of those divergences, that is, in 79.7%, the prediction carries the hyphen in the second position, that is, the model leaves empty the position that the corpus fills; and of those 8,086, 6,146 — 76.0% — fall on words whose corpus tag has p in the first position. The denominator of the second rate is therefore the subset in which the model leaves the position empty, not the total of the disagreements: of the 10,141 disagreements, 6,393, that is, 63.0%, fall on pronouns. The other three combinations behave in the same way. The disagreements are 4,906 in Gorman × Stanza, 7,645 in Gorman × UDPipe and 8,004 in Pedalion × Stanza; the share of predictions with the hyphen is 82.6%, 76.6% and 87.1%; the share of pronouns within those predictions is 83.1%, 63.8% and 89.1%.

S10.7 Agreement by part of speech

The part of speech is the one written in the corpus, not the predicted one: the question is how the models behave on the words that the corpus classifies in a certain way. The classes with at least two hundred words are reported.

Table S10.6 Agreement by part of speech of the corpus, Gorman. The denominator of the lemma and that of the tag differ slightly because they exclude different sets of words (S10.2).

Part of speech	Words (lemma)	Stanza	UDPipe	Words (tag)	Stanza	UDPipe
n noun	110,780	82.2%	76.5%	110,806	86.3%	79.6%
v verb	107,782	84.0%	63.4%	107,642	87.8%	75.8%
d adverb	79,506	92.6%	87.6%	79,476	76.7%	33.9%
l article	76,363	99.5%	97.8%	76,355	92.1%	79.8%
a adjective	54,784	87.0%	79.2%	54,717	60.0%	54.1%
c conjunction	42,328	91.2%	87.4%	42,327	88.1%	71.8%
r preposition	38,741	93.4%	88.2%	38,741	96.6%	88.4%
p pronoun	37,488	93.7%	92.1%	37,488	71.7%	36.0%
i interjection	922	98.9%	98.8%	922	98.5%	98.5%

Table S10.7 Agreement by part of speech of the corpus, Pedalion. The row of the local extension b shows that no model can write a value that its inventory does not contain: on the 8,555 words tagged b agreement on the tag is therefore zero by construction, whereas on the lemma the 8,556 words of this part of speech — one more, whose tag has eight characters — stand at 89.5% with Stanza and 85.8% with UDPipe.

Part of speech	Words (lemma)	Stanza	UDPipe	Words (tag)	Stanza	UDPipe
v verb	50,556	78.1%	58.5%	50,555	85.4%	71.9%
n noun	43,887	78.4%	72.8%	43,887	79.3%	70.8%
d adverb	32,125	88.9%	85.5%	32,063	65.2%	37.7%
l article	27,094	98.8%	96.5%	27,094	88.6%	73.6%
p pronoun	24,400	93.4%	90.1%	24,400	38.5%	18.4%
a adjective	20,596	81.2%	73.8%	20,596	65.8%	54.6%
c conjunction	15,224	92.2%	91.3%	15,221	71.9%	69.8%
r preposition	13,655	91.9%	87.9%	13,655	90.8%	85.3%
b local extension	8,556	89.5%	85.8%	8,555	0.0%	0.0%
m numeral	1,293	84.1%	81.0%	1,282	6.3%	22.6%
i interjection	990	91.6%	91.1%	990	80.4%	81.0%

S10.8 The three-way comparison

For each eligible word there are five mutually exclusive outcomes (§ 7.4 of the article; definitions in S10.3): all three agree; corpus and Stanza agree against UDPipe; corpus and UDPipe agree against Stanza; the two models agree against the corpus; no pair agrees. Table S10.8 gives their counts on the lemma and on the tag.

Table S10.8 Corpus, Stanza and UDPipe on the same words, with the denominators of Table S10.2. The sum of the first two rows of each column is the agreement of Stanza; the sum of the first and the third is that of UDPipe.

Outcome	Gorman lemma	%	Gorman tag	%	Pedalion lemma	%	Pedalion tag	%
all three agree	435,830	79.41	348,446	63.52	180,848	75.86	129,645	54.40
corpus and Stanza against UDPipe	53,897	9.82	108,278	19.74	24,102	10.11	40,404	16.95
corpus and UDPipe against Stanza	11,737	2.14	18,025	3.29	6,673	2.80	8,940	3.75
the two models against the corpus	18,581	3.39	43,317	7.90	9,432	3.96	33,754	14.16
no pair agrees	28,758	5.24	30,527	5.56	17,334	7.27	25,568	10.73
Total	548,803	100	548,593	100	238,389	100	238,311	100

The comparison key used in this table. The three-way comparison on the lemma adopts the same normalised key as the pairwise measures: the composed form (NFC) stripped of the Perseus lemma numbering, applied to all three sides. If the raw strings were compared instead, the fourth row would amount to 27,566 words in Gorman and 9,803 in Pedalion. In Gorman the difference from the 18,581 reported here is due entirely to the numeral appended to the lemma: 8,985 words in which corpus and models choose the same lexical entry and differ only in the final digit. In Pedalion the difference amounts to 371 words and is made up of 372 cases that the raw comparison adds — 229 due to the numbering and 143 due to a different Unicode composition of two spellings that look identical on screen — minus one case that only the normalised comparison recognises, where the corpus writes κύων and the two models κυνός. Since that digit is tool metadata and not an annotation choice, and since the pairwise measures normalise it (S10.3), the normalised comparison is the methodologically relevant one, and it is the only one used in this section.

S10.9 The recurrent substitutions, broken down by lemma

The tag substitutions on which the two models agree against the corpus — 43,317 words in Gorman and 33,754 in Pedalion (Table S10.8) — are concentrated in a few frequent patterns (§ 7.5 of the article). At the level of the complete tag, three substitutions in Gorman and four in Pedalion recur more than 1,000 times and together cover 23.3% and 34.4% of the cases, about a quarter and a third; the two most frequent in each corpus concern uninflected words: in Gorman the adverb d-------- replaced by the conjunction c-------- (6,142 words, 28 forms) and by the particle g-------- (2,735 words, 46 forms); in Pedalion the local extension b-------- replaced by c-------- (4,991 words, 16 forms) and the adverb d-------- by g-------- (4,113 words, 61 forms). Grouped by part of speech — the value of the first position in the corpus and in the models — twelve pairs recur at least 1,000 times in Gorman and together cover 92.1% of the cases: d→c 6,468, a→a 5,738, v→v 5,153, p→p 4,155, n→n 3,589, l→l 2,795, d→g 2,778, n→a 2,387, d→d 2,145, p→a 1,751, a→n 1,643 and a→p 1,313; in Pedalion eleven pairs cover 86.5%: p→p 6,445, b→c 4,991, d→g 4,113, p→a 3,258, v→v 2,082, n→n 1,810, a→a 1,803, d→c 1,413, c→d 1,129, l→l 1,121 and a→n 1,026. In Gorman the pair d→g (2,778) exceeds the substitution of the complete tag d-------- by g-------- (2,735) by 43

words, whose corpus tag is d-------_, with an underscore at the ninth position. A pair such as a→a gathers the substitutions that keep the part of speech and change other positions. Table S10.9 breaks down by lemma the two most frequent substitutions of the complete tag in each corpus; it lists only the main lemmas of each substitution, so that its rows do not add up to the totals given here.

Table S10.9 The two most frequent substitutions of the complete tag in each corpus, among the words on which the two models agree against the corpus, broken down by lemma. The column "Words of the lemma" counts the words whose normalised lemma key — the NFC composed form stripped of the Perseus lemma numbering, as in S10.8 — is the one indicated; it is not the count of a graphic form. The universe is that of the words. The last column says how many times the corpus writes, on the same lemma, precisely the tag that the models propose: it says how often the value proposed by the models is already in use in the corpus for that lemma. A high share means that the corpus writes both values on the same lemma; on its own it does not establish whether it does so to distinguish two different functions or because of internal inconsistency, and the table does not decide between the two readings.

Corpus	The corpus writes	The models write	Lemma	Words	Words of the lemma	The corpus already writes it this way
Gorman	d-------- adverb	c-------- conjunction	καί	4,543	29,574	18,801 (63.6%)
Gorman	d--------	c--------	ὡς	503	3,539	2,475 (69.9%)
Gorman	d--------	c--------	ἤ	470	1,618	1,117 (69.0%)
Gorman	d--------	c--------	μή	412	2,520	166 (6.6%)
Gorman	d-------- adverb	g-------- particle	γάρ	600	4,251	0 (0.0%)
Gorman	d--------	g--------	ἄν	508	2,210	0 (0.0%)
Gorman	d--------	g--------	δέ	436	20,088	0 (0.0%)
Gorman	d--------	g--------	μέν	307	7,278	0 (0.0%)
Pedalion	b-------- extension	c-------- conjunction	καί	4,437	12,296	5,264 (42.8%)
Pedalion	b--------	c--------	ἤ	289	1,200	892 (74.3%)
Pedalion	b--------	c--------	ἀλλά	192	1,521	713 (46.9%)
Pedalion	d-------- adverb	g-------- particle	γάρ	989	2,265	0 (0.0%)
Pedalion	d--------	g--------	δέ	825	6,847	0 (0.0%)
Pedalion	d--------	g--------	ἄν	715	1,301	0 (0.0%)
Pedalion	d--------	g--------	μέν	477	2,311	0 (0.0%)

The particle value is used by neither of the two corpora: zero times out of 548,912 words in Gorman and zero out of 238,413 in Pedalion. The models write it — among all the words, Stanza 4,418 times on Gorman and 6,131 on Pedalion, UDPipe 38,438 and 16,966 — and, among the words with a nine-character tag, they agree on it 3,013 and 4,726 times, of which 2,735 and 4,113 where the corpus writes the adverb d--------. The difference between the two models on this value, 4,418 against 38,438, is itself a piece of evidence: the two systems did not learn the same thing from the same tradition. The AGDT guidelines are themselves ambivalent on this point: those of version 1.1 treat the particle as "a morphological category for uninflected function words (like μέν, γάρ and δέ)" (Bamman and Crane 2008: 38), whereas those of version 2.0 keep g among the values of the first position but prescribe that sentence adverbs "should be tagged as adverbs at the morphological level" (Celano 2014, § 2.6). The AGDT 2.1 data on which both models are trained, in their conversion UD_Ancient_Greek-Perseus (UD_Ancient_Greek-Perseus README), use both values for the same lemmas — in its training set, in release 2.4 on which the UDPipe model was trained, g occurs on 13,629 of 159,895 tokens, and δέ receives g 6,037 times and d 1,140 times, γάρ 962 and 275 times, μέν 869 and 501 times (from release 2.11 two occurrences of δέ pass from g to d) — and the README of those data acknowledges that "Some work is still needed in order to satisfactorily

deal with the distinction adverb/particle" (AGDT 2.1 README). Gorman and Pedalion, which never use g, are thus consistent with the 2.0 prescription.

In Pedalion the lemma καί, among the words, is tagged c-------- 5,264 times and b-------- 5,135 times (one more occurrence carries b in a tag of eight characters): it is on the latter that the 4,437 substitutions of b by c on καί fall, the first Pedalion row of Table S10.9, out of the 4,991 that this substitution counts across 16 forms. That the same lemma is tagged c-------- 5,264 times and b-------- 5,135 times is compatible with two readings, and these figures do not allow a choice between them: it may be a distinction that the corpus draws and the models do not reproduce, or an internal inconsistency. The README of the Pedalion Trees declares b to be an experimental value that in most cases is equivalent to d or c (S7.4), which makes the first reading possible without proving it.

S10.10 The lemma: inside the cases of agreement against the corpus

The 18,581 cases in Gorman and the 9,432 in Pedalion were classified with formal criteria, applied in this order and mutually exclusive: (1) the two citation forms differ only in diacritics, capitalisation or final sigma, that is, they coincide once decomposed into NFD, stripped of the combining marks, lower-cased and with the final sigma reduced to σ; (2) the two models assign the word the same part of speech — the first position of the tag — and this differs from the one written in the corpus; (3) one of the two citation forms, reduced as in (1), is a prefix of the other; (4) everything else. The elision mark — U+1FBD, U+02BC or U+2019 — is not a combining mark and remains in the compared string. The fourth category is operational and residual: the row "residual" of Table S10.10 collects by definition the cases that the first three criteria did not capture, and its name says where the case ends up, not what has been established about the Greek. It is therefore not a measure of how many times corpus and models read the sentence differently: it is the complement of the three formal criteria, and its size would change if a fourth were added.

Table S10.10 Formal classification of the cases in which the two models agree with each other against the corpus, on the lemma.

Category	Gorman	%	Pedalion	%
different spelling of the citation form: diacritics, capitalisation or final sigma	3,872	20.8	1,169	12.4
different part of speech in the corpus and in the models	2,092	11.3	1,574	16.7
one citation form a prefix of the other, with diacritics removed	1,334	7.2	584	6.2
residual: none of the three formal criteria	11,283	60.7	6,105	64.7
Total	18,581	100.0	9,432	100.0

Examples. ᾗ → ἧ (611 cases in Gorman) falls entirely in the first category, because the two citation forms coincide once the accent is removed. ἄρχων → ἄρχω (108 cases in Gorman, 90 in Pedalion) falls almost entirely in the second category (104 and 82 cases): the corpus annotates the noun ἄρχων, "archon, ruler", the two models the participle of ἄρχω. ὅς → ὁ (513 cases in Gorman, 202 in Pedalion) falls in the second category when both models give the word the part of speech l, article, against the p, pronoun, of the corpus (80 and 11 cases), and otherwise in the third, because ὁ

without diacritics is a prefix of ὅς: in 261 and 81 cases the two models disagree with each other on the part of speech, in 172 and 110 both write p, as the corpus does. Criterion (3) captures truncated citation forms without an elision mark, such as ἐπειδή → ἐπεί, οὔτε → οὐ and Σύλλας → Σύλλα; elided citation forms, such as ἀλλ᾽ for ἀλλά (1,100 cases in Gorman and 419 in Pedalion, with the elision mark U+1FBD, U+02BC or U+2019), keep the elision mark and fall in the fourth category. Besides divergences of analysis, the fourth category therefore contains elided citation forms — in Gorman 2,862 of its 11,283 cases and in Pedalion 1,139 of its 6,105 have a model lemma ending with an elision mark, and 7 and 12 one beginning with it (prodelision or aphaeresis, as ᾽θέλω for ἐθέλω) — and conventions in the choice of the citation lemma, such as the suppletive λέγω → εἶπον (633 cases in Pedalion). Table S10.11 gives the category of each of the twenty most frequent substitutions.

Table S10.11 The ten most frequent lemma substitutions in each corpus, among the cases in which the two models agree against the corpus. The column “Cat.” gives the category of Table S10.10 into which the cases of the row fall; when they are split between two categories, the number in brackets says how many fall into each. Substitutions that differ only in the character of the elision mark — U+1FBD in ἀλλ᾽, U+02BC in ἀλλ’, U+2019 in ἀλλ’ — are different strings and are listed separately.

Gorman: the corpus → the models	Words	Cat.	Pedalion: the corpus → the models	Words	Cat.
ἀλλά → ἀλλ’	651	4	λέγω → εἶπον	633	4
ἢ → ἤ	611	1	δέω → δεῖ	311	4
ὅς → ὁ	513	2 (80) · 3 (433)	ἀλλά → ἀλλ’	293	4
ἀλλά → ἀλλ’	395	4	ὁράω → εἶδον	227	4
δέ → -δέ	386	2 (79) · 4 (307)	ὅς → ὁ	202	2 (11) · 3 (191)
εἰς → ἐν	374	2 (1) · 4 (373)	μάλα → μάλιστα	159	4
τις → τίς	361	1	μάλα → μᾶλλον	153	4
διά → δι’	293	4	μόνον → μόνος	152	2 (148) · 4 (4)
ἐάν → ἄν	240	2 (42) · 4 (198)	ἀλλά → ἀλλ’	125	4
μη → μή	167	1	λέγω → ἐρῶ	106	4

Among the substitutions of Table S10.11 three kinds recur, which cut across the categories of Table S10.10. The first is the elided citation form returned as the lemma (ἀλλ’, δι’), which is a choice of the tool and not a reading of the Greek. The second is the suppletive verb: Pedalion traces the aorist back to the present (λέγω for εἶπον, ὁράω for εἶδον), whereas the models keep the separate citation form of the lexicographic tradition. The third is the pair of lemmas close in form and distinguished by accent or by category (ἢ and ἤ, τις and τίς, ὅς and ὁ), where the choice depends on the analysis of the sentence.

S10.11 What is left out of the comparison, and why

Syntactic relations. They are not compared (the comparison concerns lemma and XPOS only). The two analysers used produce lemma and morphology in the AGDT positional scheme, but not a dependency relation in the AGDT inventory, and the comparison files saved by the runs do not contain a column of predicted relations. A syntactic comparison would require a parser trained on that inventory and an explicit decision on how to treat elliptic nodes — tokens with the attribute artificial or a placeholder form of the type [n] — which number 8,530 in Gorman and 4,914 in Pedalion and have no position in the text. They must not be confused with the technical tokens, which in the two corpora number 8,530 and 6,636 and also include lacunae, z markers and empty tokens (§ 3.1 of the article).

GLAUx. The four strata of GLAUx have no comparison phase and do not enter the comparison of § 7 of the article.

Other models. The comparison uses the two analysers that the program can run locally and whose resources are identified by a fingerprint. We did not compare models accessible only over the network, because their version cannot be verified at a later date and the comparison would not be reproducible.

Words without a lemma or without a tag. They remain outside the denominator of the corresponding field and are counted in Table S10.2: 109 and 24 words without a lemma, 22 and 12 without a tag, 297 and 90 with a tag whose length is not nine. Their exclusion follows from the definitions of S10.3, which the program applies when it writes the eligibility flags of the comparison tables (S10.12).

S10.12 How the predictions were obtained

The models receive the tokens supplied to the models of Table S10.2: the active tokens, punctuation included, in the order of the file and divided into the sentences of the corpus; in Pedalion one active token with an empty form is left out. Neither model receives the lemma, the morphological tag or the relation of the corpus. Each prediction is aligned with the corpus token by the quadruple of source file, document, sentence index and token index, and saved in a comparison table, one per corpus and model, in the folder of the run (S11.4): the table puts side by side the columns of the corpus and those of the model, the normalised lemma keys of both, the two eligibility flags (eleggibile_lemma, eleggibile_postag) and the two agreement flags (accordo_lemma, accordo_postag) defined in S10.3. The measures are computed on the words eligible for each field, which are the denominators of Table S10.2.

S10.13 How to recompute the figures of this section

The measures of S10.4–S10.11 are computed from the four comparison tables, with the definitions of S10.3 and the denominators of S10.2, without running the models again. Table S10.8 is obtained by joining the two tables of a corpus on the quadruple of source file, document, sentence index and token index, keeping the words eligible for the field and classifying each word by the three relations of S10.3: the agreement of corpus and Stanza and of corpus and UDPipe is the one recorded in the columns accordo_lemma and accordo_postag; the agreement of the two models is the equality of lemma_modello_normalizzato, or of postag_modello when both predictions are interpretable. The function three_way_agreement of the module modelcmp.py, which draws the corresponding chart in the interface of the program, uses another definition — raw strings, all the aligned words and the XML identifiers as key — and gives different counts (for the tag in Gorman, 43,487 cases of agreement of the models against the corpus out of 548,912 words): it is not the procedure of Table S10.8. To produce the predictions again, the two models must be run on the tokens supplied to the models (S10.12), with the versions of Table S10.1 and the model files of Table S11.3: the corpus fingerprint verifies the tokens up to Unicode normalisation, the resource fingerprint the names and sizes of the model files, and the SHA-256 of Table S11.3 their content.

S10.14 Overlap with the training data of the two models

With five-word matching, 39,065 of the 548,912 words of Gorman (7.1%) overlap with the training set of UD_Ancient_Greek-Perseus, 19,695 with its development set and 35,313 with its test set; in

the 36,632 words whose whole sentence coincides with a sentence of the training set, the morphological tag of Gorman is identical to the XPOS of UD Perseus in 98.2% of cases. In Pedalion the overlap with the training set is 1,282 words (0.5%) (§ 7.3 of the article). This section gives the reason for the measure, the data, the method, the overlap by work (Table S10.12) and its effect on the agreement rates (Table S10.13).

Both models are trained on UD_Ancient_Greek-Perseus (S10.1), an automatic conversion of passages of AGDT 2.1 (UD_Ancient_Greek-Perseus README), and the README of AGDT 2.1 lists V. Gorman among its annotators (AGDT 2.1 README). Texts of the evaluated corpora may therefore belong to the data on which the models were trained, which would favour agreement on those texts. This section measures that overlap and its effect on the measures of § 7 of the article.

Data. The training, development and test sets of UD_Ancient_Greek-Perseus contain 11,476, 1,137 and 1,306 sentences. In every release from 2.4 to 2.18 the sentence identifiers, their distribution among the three sets, their order and the word forms of every sentence are identical (each release tag of the repository of the treebank was compared, form by form); between releases only heads, relations, UPOS and two XPOS values change, and no token changes its UPOS to or from PUNCT, so that the sequences of UD tokens other than punctuation used below are also the same in every release. The forms were taken from release 2.4, on which the UDPipe model was trained (S10.1); the release used to train the Stanza package is not documented, but since the forms are identical in all those releases the measure of overlap does not depend on it. The corpus side consists of the words of the comparison tables of the two runs of 9 September 2026 (S10.1, S10.2).

Method. The forms of the corpus words and of the UD tokens other than punctuation (UPOS other than PUNCT) are normalised in the same way: NFC composition, a single character for every sign used as an elision mark (U+2019, U+02BC, U+1FBD, U+1FBF, U+0313, U+2018 and the ASCII apostrophe), grave accent turned into acute, lower case. A word of the corpus overlaps with one of the three UD sets when it falls within a sequence of five consecutive words of its sentence that also occurs, in the same order, in a sentence of that set; a word that overlaps with more than one set is assigned first to the training set, then to the development set and then to the test set. Sentences that coincide as a whole after normalisation are counted separately. The agreement measures are those of S10.3, computed on the same words.

Gorman. Besides the 39,065 words that overlap with the training set (7.1%), 19,695 overlap with the development set and 35,313 with the test set; 454,839 (82.9%) do not overlap. The overlap is concentrated in a few works (Table S10.12): Herodotus 1 and Diodorus 11 in the training set; Athenaeus 12–13 mostly in the test set and in part in the training set; Thucydides 1 and Plutarch's *Alcibiades* and *Lycurgus* in the development set. In the other works the matches are isolated sequences: 259 words in all for the training set. Sixteen of the eighteen Athenaeus sections of Gorman are present in two versions, as separate files (34 files in all, with names ending in -2019 and -jan-15), so that most Athenaeus words are counted twice, as in all the counts of the corpus. In the 36,632 words whose whole sentence coincides with a sentence of the training set, the morphological tag of Gorman is identical to the XPOS of UD_Ancient_Greek-Perseus in 35,960 cases (98.2%) and the lemma, without the Perseus numbering, in 99.7%.

Pedalion. 1,282 of its 238,413 words (0.5%) overlap with the training set, 257 with the development set and 119 with the test set, mostly in the files of example sentences

(external_examplesentences.xml, chilia-sentences.xml and example-sentences.xml); 1,097 words belong to sentences that coincide with sentences of the training set.

Table S10.12 The words of Gorman and Pedalion by overlap with the three sets of UD_Ancient_Greek-Perseus, by work. A word is assigned to the first set, in the order training, development, test, with which it shares a sequence of five words. Athenaeus 12 and 13 comprise both versions of the files present twice in Gorman.

Corpus	Work (files)	Training	Development	Test	No overlap	Words
Gorman	Herodotus 1 (hdt-1)	17,095	0	0	12,081	29,176
Gorman	Diodorus 11 (diodsic)	15,242	0	0	7,831	23,073
Gorman	Athenaeus 12 (athen12)	212	1,525	15,757	13,968	31,462
Gorman	Athenaeus 13 (athen13)	6,247	0	19,519	17,749	43,515
Gorman	Thucydides 1 (thuc-1)	5	9,229	0	13,240	22,474
Gorman	Plutarch, Alcibiades (plut-alcib)	5	4,343	5	5,727	10,080
Gorman	Plutarch, Lycurgus (plutarch-lycurgus)	0	4,451	0	4,933	9,384
Gorman	other works	259	147	32	379,310	379,748
Gorman	total	39,065	19,695	35,313	454,839	548,912
Pedalion	total	1,282	257	119	236,755	238,413

Effect on the agreement. Table S10.13 gives the rates of S10.3 by overlap class. On the words that overlap with the training set Gorman agrees with the models much more than on the words without overlap; excluding those words lowers the four rates of Gorman by 0.6 to 1.4 points and those of Pedalion by less than 0.1. The share of words on which the two models agree with each other against the corpus on the tag is 2.2% on the training overlap of Gorman and 8.4% on its words without overlap; without the training overlap it rises from 7.9% to 8.3% in Gorman and stays at 14.2% in Pedalion. The overlap therefore inflates the agreement on Gorman slightly, but it does not explain the difference between the two corpora or the cases of agreement against the corpus discussed in § 7 of the article.

Table S10.13 Agreement by overlap class: numerator, eligible words and rate, with the definitions of S10.3. “All words” gives the measures of Table S10.3; “without training overlap” excludes the words that overlap with the training set. The last measure of each corpus is the share of words on which the two models agree with each other against the corpus on the tag.

Corpus	Measure	Overlap class	Agreements	Eligible words	Rate
Gorman	lemma, Stanza	training	38,110	39,063	97.56%
Gorman	lemma, Stanza	development	17,721	19,695	89.98%
Gorman	lemma, Stanza	test	30,220	35,313	85.58%
Gorman	lemma, Stanza	no overlap	403,676	454,732	88.77%
Gorman	lemma, Stanza	all words	489,727	548,803	89.24%
Gorman	lemma, Stanza	without training overlap	451,617	509,740	88.60%
Gorman	lemma, UDPipe	training	38,160	39,063	97.69%
Gorman	lemma, UDPipe	development	16,079	19,695	81.64%
Gorman	lemma, UDPipe	test	27,533	35,313	77.97%
Gorman	lemma, UDPipe	no overlap	365,795	454,732	80.44%
Gorman	lemma, UDPipe	all words	447,567	548,803	81.55%
Gorman	lemma, UDPipe	without training overlap	409,407	509,740	80.32%
Gorman	tag, Stanza	training	35,731	39,062	91.47%
Gorman	tag, Stanza	development	16,943	19,694	86.03%
Gorman	tag, Stanza	test	28,886	35,165	82.14%
Gorman	tag, Stanza	no overlap	375,164	454,672	82.51%
Gorman	tag, Stanza	all words	456,724	548,593	83.25%
Gorman	tag, Stanza	without training overlap	420,993	509,531	82.62%
Gorman	tag, UDPipe	training	33,108	39,062	84.76%
Gorman	tag, UDPipe	development	13,404	19,694	68.06%

Corpus	Measure	Overlap class	Agreements	Eligible words	Rate
Gorman	tag, UDPipe	test	23,708	35,165	67.42%
Gorman	tag, UDPipe	no overlap	296,251	454,672	65.16%
Gorman	tag, UDPipe	all words	366,471	548,593	66.80%
Gorman	tag, UDPipe	without training overlap	333,363	509,531	65.43%
Gorman	tag, both models against the corpus	training	873	39,062	2.23%
Gorman	tag, both models against the corpus	development	1,322	19,694	6.71%
Gorman	tag, both models against the corpus	test	2,897	35,165	8.24%
Gorman	tag, both models against the corpus	no overlap	38,225	454,672	8.41%
Gorman	tag, both models against the corpus	all words	43,317	548,593	7.90%
Gorman	tag, both models against the corpus	without training overlap	42,444	509,531	8.33%
Pedalion	lemma, Stanza	training	1,178	1,282	91.89%
Pedalion	lemma, Stanza	development	224	257	87.16%
Pedalion	lemma, Stanza	test	102	119	85.71%
Pedalion	lemma, Stanza	no overlap	203,446	236,731	85.94%
Pedalion	lemma, Stanza	all words	204,950	238,389	85.97%
Pedalion	lemma, Stanza	without training overlap	203,772	237,107	85.94%
Pedalion	lemma, UDPipe	training	1,131	1,282	88.22%
Pedalion	lemma, UDPipe	development	215	257	83.66%
Pedalion	lemma, UDPipe	test	100	119	84.03%
Pedalion	lemma, UDPipe	no overlap	186,075	236,731	78.60%
Pedalion	lemma, UDPipe	all words	187,521	238,389	78.66%
Pedalion	lemma, UDPipe	without training overlap	186,390	237,107	78.61%
Pedalion	tag, Stanza	training	939	1,282	73.24%
Pedalion	tag, Stanza	development	180	257	70.04%
Pedalion	tag, Stanza	test	97	119	81.51%
Pedalion	tag, Stanza	no overlap	168,833	236,653	71.34%
Pedalion	tag, Stanza	all words	170,049	238,311	71.36%
Pedalion	tag, Stanza	without training overlap	169,110	237,029	71.35%
Pedalion	tag, UDPipe	training	892	1,282	69.58%
Pedalion	tag, UDPipe	development	135	257	52.53%
Pedalion	tag, UDPipe	test	84	119	70.59%
Pedalion	tag, UDPipe	no overlap	137,474	236,653	58.09%
Pedalion	tag, UDPipe	all words	138,585	238,311	58.15%
Pedalion	tag, UDPipe	without training overlap	137,693	237,029	58.09%
Pedalion	tag, both models against the corpus	training	202	1,282	15.76%
Pedalion	tag, both models against the corpus	development	54	257	21.01%
Pedalion	tag, both models against the corpus	test	10	119	8.40%
Pedalion	tag, both models against the corpus	no overlap	33,488	236,653	14.15%
Pedalion	tag, both models against the corpus	all words	33,754	238,311	14.16%
Pedalion	tag, both models against the corpus	without training overlap	33,552	237,029	14.16%

Limits. The measure identifies shared text, not shared annotation: a passage present in both resources may have been revised independently in each. The five-word window also counts short sequences that different works happen to share. The development and test sets are not training data; their overlap is reported for completeness. The measure does not cover other material used by the models, such as the pretrained word vectors of the Stanza package (conll17). The figures can be recomputed from the comparison tables (S10.12) and from the CoNLL-U files of UD_Ancient_Greek-Perseus.

S11. How to recompute the figures

The results of the article come from six runs of Greek Corpus Pipeline 3.1.0, one per stratum (Table S11.1). Each run performs a preparation phase and a diagnosis phase; the runs on Gorman and on Pedalion also perform two comparison phases, one per model, which produced the four

combinations of corpus and model of § 7 of the article (S10.1). This section gives the counting units (S11.1), the frozen results (S11.2), the fingerprints (S11.3), the licences, versions and identification of the corpora (S11.4) and what is needed to repeat the measurements (S11.5).

S11.1 The five counting units

The module src/greek_corpus_pipeline/units.py defines five counting units, and every number shown by the interface declares which unit it uses. The definitions below are the operational ones. The first four coincide with those of § 3.1 of the article; the fifth, analysable sentences, is a unit of the program that the article does not use, and it must not be confused with the "sentences with at least one evaluated arc" of Table 2 of the article.

Imported tokens. The word elements of the XML files whose identifier is an integer, punctuation and technical nodes included; elements whose identifier is not an integer are not imported. A token is thus every word element that GCP imports, whether it is a word of the text, a punctuation mark or a technical node.

Active tokens. The imported tokens that remain after the adopted filter configuration. In the study the filter excludes the technical tokens and excludes neither punctuation nor non-Greek forms. A token is technical when it presents at least one of five criteria, each sufficient on its own: the attribute artificial filled in; a placeholder form consisting of a number in square brackets, such as [0] (the filter accepts any number; the largest in the data is [16]); the lacuna form G or G? (the filter also accepts a lower-case g, which does not occur); form and morphological tag both empty; a morphological tag beginning with z. These are the five criteria stated in § 3.1 of the article. They identify different types of technical elements and do not imply that all of them are elliptic nodes: recognising them is preprocessing, not diagnosis, and none of them produces a finding.

Words. The imported tokens that are neither technical nor punctuation, as in § 3.1 of the article. Being a word is a property of the token and does not depend on the filter: if the exclusion of technical tokens is switched off, the active tokens include them, but the words still do not. With the filter adopted in the study, the words are therefore the active tokens minus the punctuation among them: 616,646 − 67,734 = 548,912 in Gorman and 274,609 − 36,196 = 238,413 in Pedalion. The row "tokens classified as punctuation" of Table 2 of the article counts instead the punctuation among all the imported tokens, technical ones included (67,790 and 37,638), and cannot be subtracted from the active tokens.

The punctuation predicate has two tests, and the morphological tag takes precedence: the relation decides only when the tag says nothing. (1) The tag begins with u: the token is punctuation, whatever its relation. (2) The relation is AuxG, AuxK or AuxX and the tag says nothing usable: it does not have nine characters, or its first position carries no interpretable value of the inventory, the documented local extensions such as b counting as interpretable. A token that is not technical and whose tag has nine characters and an interpretable first position other than u is therefore a word, whatever relation it carries: a Greek word to which a punctuation relation has been assigned by mistake stays among the words.

Analysable links. The dependencies that start from an active token and have as their head an active token of the same sentence. In practice: the value of head must be greater than zero, and the pair formed by the sentence key and that value must appear among the pairs of sentence key and identifier of the active tokens. Hence no analysable link is produced by tokens with head equal to 0,

because zero is the root and not another token; by those with an empty or non-numeric head; or by those whose head, although it exists in the file, does not appear among the active tokens of the sentence — these are largely arcs pointing to an elliptic node parked after the full stop, whose position in the file is not that of the element understood in the text. The definition does not require head and dependent to be different tokens.

Analysable sentences. The sentences that have a usable structure, according to two joint requirements: a single root that is not structural punctuation, and no arc pointing to an identifier absent from the sentence. The predicate is computed on the imported tokens, and a root is a token with head 0 whose relation is not AuxG, AuxK or AuxX; technical tokens count, so that an elliptic node attached to 0 can be the single root of a sentence. The first requirement excludes structural punctuation because in AGDT a punctuation mark can be attached to 0 without thereby being a second root: Gorman has 88,692 tokens with head 0 in 26,616 sentences, and 27,305 roots once the punctuation so defined is excluded. The punctuation to be excluded here is the one defined by the relation alone — AuxG, AuxK, AuxX — and not the one of the predicate used for words: a comma that carries COORD is a node of the tree and can be its root. The same definition of root is used by the checks sentence-no-root and sentence-many-roots (S2).

This unit does not coincide with the “sentences with at least one evaluated arc” of Table 2 of the article, which are the denominator of the sentence-level measure of non-projectivity. The two count different sets and give different numbers: the analysable sentences are 25,994 in Gorman and 18,235 in Pedalion, the sentences with at least one evaluated arc 26,377 and 17,984. The first unit requires a single non-punctuation root and no non-existent head, and a sentence with a single broken arc drops out of it entirely; the second requires only that the sentence contain at least one arc that entered the projectivity structure, and a sentence with one broken arc and twenty good ones falls within it. The first unit does not test acyclicity or connectivity: among the analysable sentences, 16 in Gorman and 28 in Pedalion have tokens whose chain of heads does not reach the root, and in nine of the Pedalion ones the chain contains a cycle (in eight when the technical tokens are left out). The two names are not synonyms and must not be interchanged.

The projectivity structure. In order to verify projectivity, GCP builds a separate structure of arcs, excluding an arc if it involves a node with the attribute artificial, a placeholder form of the type [n] or a node placed after the last sign marked AuxK. The last criterion is an operational criterion specific to GCP, not a general definition of an elliptic node nor a rule of AGDT, and in the data analysed the small difference between analysable links and links covered by the projectivity structure (Table 2 of the article: 513,369 and 513,355 in Gorman, 213,144 and 213,113 in Pedalion) depends exclusively on it; in the AGDT guidelines, AuxK identifies instead the final punctuation mark of the sentence, attached to the root (Celano 2014, § 3.16). The row “links covered by the projectivity structure” of Table 2 documents the actual coverage of that structure and is not the denominator of the percentage of non-projective links. In verifying whether a link is non-projective, GCP does not consider the punctuation among the intervening tokens; for this specific test punctuation is identified when the token has u as its part of speech or the relation AuxG, AuxK or AuxX, irrespective of the more restrictive criterion used to establish which tokens count as words. The two uses of “punctuation” must therefore be kept distinct. The file arcs.parquet of each run contains the arcs of the projectivity structure; the figures of Table 2 of the article are those of the arcs whose dependent and head are both active tokens, which in Gorman are all the arcs of the file

and in Pedalion 213,113 of its 213,549, the other 436 involving technical tokens that are not elliptic, such as lacunae and z markers.

The sentence key is not the field sentence_id, which in Gorman takes 564 distinct values over 26,616 sentences because it numbers the sentence within its own file: grouping on it would merge the sentences that share a number across 158 documents. The key is the concatenation of source file, document and sentence index.

S11.2 The frozen results

The table gives the six strata with the filter configuration adopted in the study: technical tokens excluded, punctuation and non-Greek forms included. These are the values against which to compare in order to establish whether a new run reproduces the measurement.

Table S11.1 The frozen results of the six strata. The column “Active checks” gives the number of checks that produce at least one finding in that stratum, not the number of checks applied, which is 77 throughout. The last column is the findings fingerprint, defined in S11.3.

Stratum	Imported tokens	Active tokens	Words	Analysable links	Analysable sentences	Findings	Active checks	Nuclei	Fingerprint
Gorman	625,176	616,646	548,912	513,369	25,994	101,047	56	33,652	971c8056de40
Pedalion	281,245	274,609	238,413	213,144	18,235	29,515	51	13,848	a037eae69887
GLAUx automatic	300,292	292,567	251,994	223,872	17,982	13,168	22	11,347	6560da673493
GLAUx manual	21,652	21,107	17,834	16,713	1,745	2,400	20	996	e9c9e830c5b6
GLAUx confirmation automatic	273,354	267,313	231,725	206,955	13,227	11,259	20	11,051	a3efc2fa2260
GLAUx confirmation manual	28,666	28,143	24,351	21,769	1,979	3,188	20	1,041	771c3f3dbedd

The findings are 160,577 in all, produced by the 77 checks of the catalogue. The runs on Gorman and on Pedalion also contain the comparison phases with Stanza and UDPipe documented in S10.1. The six runs are identified, with their date and the files they read, in Table S11.2 (S11.4).

The six runs were carried out on a Windows machine with Python 3.14.6, pandas 3.0.5, NumPy 2.5.3 and PyArrow 25.0.1. They have been repeated on a Linux system with the same versions, version 3.1.0 of the program, the same XML files (Table S11.2) and the same configuration: the repetition reproduces cell by cell the tokens, the findings with their proposals, the arcs and the tables of the variation nuclei of all six strata, and, with Stanza 1.14.0 (PyTorch 2.14.0), UDPipe 1.4.0.1 and the model files of Table S11.3, the four comparison tables and their evaluations.

S11.3 The four fingerprints, and what each one identifies

The word “fingerprint” is used in these materials for four different objects, which are not comparable with one another. The findings fingerprint of a stratum, the last column of Table S11.1, has twelve characters and is defined below. The corpus fingerprint and the fingerprint of the model’s resources, the two central columns of Table S10.1, have sixteen and are defined in S10.1: the former identifies the text supplied to the model, up to Unicode normalisation, the latter the names and sizes of the files of the model, whose content is identified by the SHA-256 of Table S11.3. The input fingerprint of a stratum, in Table S11.2, has sixteen and is defined in S11.4: it identifies the XML files that the run read, and for the forty texts of GLAUx Table S9.1 also gives the first twelve characters of the SHA-256 of each file. The program also computes a fifth fingerprint, used only internally to recognise the resources for which it distributes metadata (risorse-

con-metadati, S6), which is not used in these materials. The first fingerprint says what the program found, the others on what and with what it worked; neither kind implies the other.

The findings fingerprint of a stratum consists of the first twelve hexadecimal characters of the MD5 of a list of lines, one per finding, each made of four fields separated by the character U+241F: the identifier of the check, the document, the identifier of the sentence and the identifier of the token in the XML file (the attribute id of the word element). The lines are sorted as strings, duplicates included, and joined by a line break, without a final line break; the procedure is that of the function _impronta of tests/test_diagnosi.py, and it reproduces the six values of Table S11.1. The source file is not among the fields, and in the six runs this creates no ambiguity, because the document is the name of the file without its extension and corresponds to it one to one. The fingerprint identifies which check reported which position, independently of the order in which the findings are output and of the text with which they are described; it does not cover the values diagnosed, the parameters of the diagnosis, the proposed corrections or the severity, nor the descriptive columns. In the tables that a run writes, these columns are in Italian whatever the language of the interface; the interface shows them in its current language. Two runs with the same fingerprint therefore produced findings of the same checks at the same positions, not necessarily with the same proposals. Where the XML identifiers of two tokens of the same sentence coincide — in Pedalion, in seven sentences — the fingerprint cannot tell which of the two carries a finding: this concerns 17 findings.

S11.4 The three corpora: licences, versions and identification

The licences of the three resources are not uniform. Gorman Trees: repository perseids-publications/gorman-trees, folder public/xml; the TREEBANK_LICENSE file declares CC BY-SA 4.0, while the README, which refers to that same file, and Gorman (2020) declare CC0 1.0; the GLAUx README describes the author’s repository, vgorman1/Greek-Dependency-Trees, as available under CC BY-NC-SA 4.0. Pedalion Trees: repository perseids-publications/pedalion-trees, folder public/xml; the TREEBANK_LICENSE file declares CC BY-SA 4.0, and the README announces that licence as the one the project will probably assign. GLAUx: repository alekkeersmaekers/glaux, folder xml; according to its README the data are generally available under CC BY-SA 4.0, but some texts, and possibly some manual annotations, have a more restrictive licence, indicated text by text in the metadata file.

No corpus is distributed with the program, which downloads them on request: for Gorman and Pedalion the archive of the master branch of the repository, at every run; for GLAUx the files of the main branch listed in data/glaux_sample.csv, reusing those already present in its download folder. The six runs were carried out on 9 September 2026 (Table S11.2). For Gorman and Pedalion the archive was downloaded at the start of the run; its comment records the commit of the master branch, and all its files carry the date of that commit, 20 January 2022: the branch received no commit between that date and the download, so that the versions read by the runs are the ones that were in the repositories on 24 July 2026, the date on which, according to § 3.1 of the article, they were verified. For GLAUx the program records neither the date of the download nor the commit; the forty files of its download folder carry the date of the runs, 9 September 2026, and are identified by their SHA-256, of which Table S9.1 gives the first twelve characters, and the DOI of GLAUx, 10.5281/zenodo.10948374, designates the whole series of its versions. These resources continue to be revised, and a version downloaded today may differ from the one measured here: Table S11.2 identifies the files read by each run, and Table S11.3 the commits and archives of Gorman and

Pedalion and the model files of the comparison. A fingerprint makes it possible to verify the identity of a copy, not to recover a version that is no longer available. The files that the runs write into their folders in workspace/runs — the imported tokens, the arcs, the findings, the variation nuclei and the comparison tables, which reproduce the annotated text of the corpora — are not part of the source repository of the program: they are regenerated by running the program on the inputs of Table S11.2, and Table S11.1 gives the values against which to check the result (S11.5).

Table S11.2 The inputs of the six runs. The run is identified by the name of its folder in workspace/runs, which contains the date and UTC time of creation; the source is the one recorded in the run metadata; the XML files are those read by the preparation phase, with their overall size. The input fingerprint is the first sixteen hexadecimal characters of the SHA-256 of a list of lines “file name:SHA-256 of the file”, one per XML file, sorted and joined by a line break; the six fingerprints and the sizes were recomputed on the files of the download folder of the runs, and the imported tokens, counted file by file, match Table S11.1.

Stratum	Run	Source	XML files	Bytes	Input fingerprint	Imported tokens
Gorman	20260909-053400-gorman	gorman-trees, master branch: archive downloaded at the start of the run (2026-09-09 05:34 UTC)	158	70,735,878	dad905a69a0848cb	625,176
Pedalion	20260909-060742-pedalion	pedalion-trees, master branch: archive downloaded at the start of the run (2026-09-09 06:07 UTC)	40	40,811,672	1b799d2b82395412	281,245
GLAUx automatic	20260909-064439-glaux-campione	glaux, main branch: the 18 files of data/glaux_sample.csv, reused if already downloaded	18	47,167,010	fcbad2724b8b6290	300,292
GLAUx manual	20260909-070338-glaux-campione-manuale	glaux, main branch: the 2 files of data/glaux_sample.csv, reused if already downloaded	2	3,580,782	ddcfd293d1908680	21,652
GLAUx confirmation automatic	20260909-065606-glaux-conferma	glaux, main branch: the 17 files of data/glaux_sample.csv, reused if already downloaded	17	44,514,557	b03f5a325a2a8e6c	273,354
GLAUx confirmation manual	20260909-070407-glaux-conferma-manuale	glaux, main branch: the 3 files of data/glaux_sample.csv, reused if already downloaded	3	4,532,597	a4ac3b46084fdcaf	28,666

Table S11.3 The identifiers of the archives of Gorman and Pedalion and of the model files of the comparison (S10.1). For the two archives the commit is the one written in the comment of the ZIP archive downloaded by the program and the SHA-256 is that of the archive as it was downloaded; for the model files the SHA-256 is that of the files in the model folder of the program (workspace/models), on which the two resource fingerprints of Table S10.1 were recomputed with the same result.

Object	Identifier
gorman-trees, master branch	commit 83d93a554a46ff6e9a34a15a9e0be2dff270a612
archive gorman-trees-master.zip	SHA-256 89285eae23ba026d23db417b461f9f021489efa94a079f042883a559f80a9592
pedalion-trees, master branch	commit 112c106b86b865ddde4be7e5f3e409d9c42689f9
archive pedalion-trees-master.zip	SHA-256 d7ed7ccdeb5aef09ed0f5c509a7e9fce264dc5a93bf59799b8de6c0fc1e69973
Stanza: resources.json (457,371 bytes)	SHA-256 4e41c1df152146fa26ed0c006a08feea7a60bb3414bb6d57dbda24ad2e3cb99c
Stanza: grc/tokenize/perseus.pt (643,313 bytes)	SHA-256 9e1cb98d39fa21dc543c1028d8d4fb6a534e2187051b15264b37498f9f06abb3
Stanza: grc/pos/perseus_nocharlm.pt (18,372,328 bytes)	SHA-256 3ede3a6c272fa9219066a6089846e6ac77fcc2ecfd729348a674cc544a5092d1
Stanza: grc/lemma/perseus_nocharlm.pt (2,595,726 bytes)	SHA-256 f905663df890b4bf418dfde1212ed9f6bf6b096c8d93dfc669d7b44d28449dfb
Stanza: grc/pretrain/conll17.pt (19,844,392 bytes)	SHA-256 3bd32434b47d7c060036da42070f9cf2709e1e9dd315d045027bc69806317807
UDPipe: ancient_greek-perseus-ud-2.4-190531.udpipe (20,492,125 bytes)	SHA-256 b85df6199730506b33cf8aa99f5a02b63bcbaaae94f78fc4c1f5f0bdd6235505

S11.5 What is needed to repeat the measurements

The program requires Python 3.11 or later and is distributed under the MIT licence with its source code, the catalogue of checks (data/catalogo_controlli.csv), the ten closed lists (data/liste) and the list of the forty GLAUx texts (data/glaux_sample.csv). Stanza and UDPipe are optional

dependencies, needed only for the comparison with the models. Every run writes into its own folder (S11.4) the imported tokens, the arcs, the findings, the variation nuclei by level and, when the models are run, the comparison tables: all the measures of the six runs and of the four comparisons can be derived from those files, which are not distributed with the program (S11.4), without running the diagnosis or the models again. The sensitivity analysis of S5 also requires the program, and the overlap of S10.14 the CoNLL-U files of UD_Ancient_Greek-Perseus.

The release of the program includes a suite of automated tests (folder tests). Most of them run on synthetic data and on hand-written fragments and pass without downloading anything; a minority require the six frozen strata, the UDPipe model or an otherwise idle machine on which to measure timings, and in their absence they are skipped with the reason stated.

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